

Simple Threshold (Fully Homomorphic) Encryption from LWE with Polynomial Modulus

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Abstract. The learning with errors (LWE) assumption is a powerful tool for building encryption schemes with useful properties, such as plausible resistance to quantum computers, or support for homomorphic computations. Despite this, essentially the only method of achieving threshold decryption in schemes based on LWE requires a modulus that is superpolynomial in the security parameter, leading to a large overhead in ciphertext sizes and computation time.

In this work, we propose a (fully homomorphic) encryption scheme that supports a simple t-out-of-n threshold decryption protocol while allowing for a polynomial modulus. The main idea is to use the Rényi divergence (as opposed to the statistical distance as in previous works) as a measure of distribution closeness. This comes with some technical obstacles, due to the difficulty of using the Rényi divergence in decisional security notions such as standard semantic security. We overcome this by constructing a threshold scheme with a weaker notion of one-way security and then showing how to transform any one-way (fully homomorphic) threshold scheme into one guaranteeing indistinguishability-based security.

1 Introduction

In a public key encryption (PKE) scheme, one needs the secret key sk to decrypt an encrypted message. Giving one single party control of the whole secret key can be seen as a single point of failure. The study of PKE with threshold decryption aims to mitigate this by splitting the secret key into n key shares $\mathsf{sk}_1, \ldots, \mathsf{sk}_n$, such that several key shares are needed to be able to decrypt ciphertexts. This is known as threshold public key encryption (ThPKE). In the common t-out-of-n setting, any set of t parties or fewer learns no information about encrypted messages, while any set of $t+1$ parties can jointly decrypt ciphertexts. To decrypt, the parties first compute their own partial decryption shares and then combine them together to recover the encrypted message. When $t = n - 1$, we call it full-threshold decryption.

Recently, NIST announced the standardization of the first cryptosystems to provide security even in the presence of quantum computers.[1] Among the finalists

[1] https://csrc.nist.gov/projects/post-quantum-cryptography.

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J. Guo and R. Steinfeld (Eds.): ASIACRYPT 2023, LNCS 14438, pp. 371–404, 2023.
https://doi.org/10.1007/978-981-99-8721-4_12

to be standardized, a majority base their security on the presumed hardness of (structured) lattice problems, such as Dilithium [Lyu+20] and Kyber [Sch+20] based on the (module) learning with errors problem (M-LWE) [LS15]. NIST also just began a project on threshold cryptography, which aims to produce guidelines and recommendations for implementing threshold cryptosystems.

It is thus a very important research question to study the possibility of thresholdizing lattice-based PKE schemes. This line of research has been initiated by [BD10], where they proposed a threshold key generation and decryption starting from Regev's encryption scheme [Reg05]. To split the secret key they use replicated secret sharing, which has a complexity that scales with $\binom{n}{t}$. Later, it has been shown that we can even build full-threshold decryption for fully homomorphic encryption (FHE) schemes [Ash+12]. A threshold fully homomorphic encryption scheme (ThFHE) allows to perform arbitrary computations on encrypted data and afterwards to partially decrypt the outcome of the computations. Their results have then been extended to t-out-of-n threshold and other access structures [Bon+18].

All works above have in common that they use a technique called *noise flooding* to guarantee that partial decryption shares do not leak any information on the underlying secret key. More precisely, each party first computes a "noiseless" partial decryption of a ciphertext using their secret key share. The noiseless partial decryptions allow recovering the message, but also reveal a small noise term e_{ct} that depends on the given ciphertext and the secret key. To prevent this leakage, every party locally adds some fresh noise on their decryption share before they jointly combine the necessary number of shares to recover the message. After decryption, the revealed noise term becomes $e_{\mathsf{ct}}+e'$, where $e' \leftarrow \mathcal{D}_{\mathsf{flood}}$ is a noise term that is hidden to the adversary. When proving security, the real partial decryption shares are replaced by simulated ones which do not depend on the secret key, and instead reveal noise terms of the form $e' \leftarrow \mathcal{D}_{\mathsf{flood}}$. By arguing that the statistical distance between both ways of deriving partial decryption shares is negligible, one can argue security. While this approach has the advantage of being rather simple, it has the drawback of requiring the ratio between the flooding noise and the size of the ciphertext noise e_{ct} to be superpolynomial in the security parameter. This in turn requires the LWE problem to be secure with a superpolynomial modulus-to-noise ratio, which weakens security and requires larger LWE parameters to compensate.

Recently, multi-party reusable non-interactive secure computation (MrNISC) was constructed from LWE with a polynomial modulus [Ben+21, Shi22]. This leads to a construction of full-threshold (multi-key) FHE with a polynomial modulus. It seems plausible that their construction can also be extended to build t-out-of-n threshold FHE with polynomial modulus; however, their techniques are very complex, due to a non-black-box "round-collapsing" technique based on garbled circuits, so unlikely to be practical. We thus started our work asking the following research question:

> Is it possible to construct a fully homomorphic encryption scheme that supports a *simple* t-out-of-n threshold decryption while allowing for a *polynomial modulus*?

Our Results. We give a positive answer to this question. On a high level, we show that the simple threshold decryption technique from previous works [BD10, Bon+18] can be significantly improved by replacing the noise flooding analysis with respect to the statistical distance by one with respect to the Rényi divergence (RD). Doing so comes with the benefit of only requiring a polynomial ratio between ciphertext noise and flooding noise, hence allowing for the desired polynomial modulus. However, it comes with several additional challenges. First, the Rényi divergence fits well in search-based security notions, such as OW-CPA security[2], but does not work well with decision-based security notions, such as the standard IND-CPA security.[3] Furthermore, it is especially difficult to apply the Rényi divergence to obtain simulation-based security, as required for typical notions of threshold decryption, since a small RD between two distributions does not imply a small statistical distance.

To overcome these challenges, we define new game-based notions of OW-CPA and IND-CPA security for threshold homomorphic cryptosystems, which are compatible with Rényi divergence-based proofs, whilst also giving desirable security guarantees for applications. Then, we give general transformations from OW-CPA to IND-CPA security for ThPKE and ThFHE schemes, and finally also show how to construct OW-CPA schemes based on the (module) LWE assumption with a polynomial modulus. For our transformation to go through, we also need the OW-CPA scheme to be *circuit private*; while this property is often achieved using noise flooding techniques that require a large modulus, it is also possible to use bootstrapping [DS16] or GSW-style FHE [Bou+16] to obtain circuit privacy with a polynomial modulus.

Put together, these techniques lead to our main result of ThFHE from (module) LWE with a polynomial modulus. More precisely, in our construction the modulus q scales as $O(\sqrt{\ell})$, where ℓ is the number of partial decryption queries made by an adversary within the security game, so q is polynomial as long as ℓ is polynomially-bounded in advance.

What About IND-CCA *Security?* We could likely upgrade our construction (for PKE) to be IND-CCA secure using non-interactive zero-knowledge proofs, similarly to [Dev+21a]. However, note that (adaptive) IND-CCA security is not possible for homomorphic encryption, and IND-CPA is still useful for standard PKE; indeed, [HV22] showed that an IND-CPA secure KEM suffices to prove security of TLS-1.3. Furthermore, when running TLS with ephemeral keys and no key re-use, the adversary only ever sees a single ciphertext under any public key—this is an ideal use-case for using our ThPKE construction in a threshold post-quantum TLS setting (e.g. for hardening security of a TLS server), since we only need to choose the parameters to be secure against a single decryption query.

[2] OW-CPA security for PKE roughly says that given the public key and an encryption of a random message m, it is hard to guess m.

[3] Unless a property called *public sampleability* is fulfilled [Bai+18].

1.1 Overview of Techniques

Defining IND-CPA *security for* ThFHE *(Sect. 3).* Most of the previous IND-CPA security definitions of ThFHE required the underlying FHE scheme to be IND-CPA secure and the partial decryptions to be statistically simulatable, e.g. [Bon+18]. When replacing the statistical distance by the Rényi divergence, however, we cannot prove the statistical simulation anymore and instead have to move to a game-based notion that combines the IND-CPA game and the partial decryption queries together into one single game. Here, to support homomorphic computations, we consider a game where in each partial decryption query, first some homomorphic evaluation is performed on a set of ciphertexts, before giving decryptions of the result to the adversary. *When* and *how* the adversary gets access to the partial decryption oracle within the IND-CPA game crucially impacts the strength of the achieved security. For example, one can allow the adversary to only query partial decryptions *before* seeing the challenge ciphertext. This was done in a previous version of ThFHE [Cho+22a], which also uses a Rényi divergence based analysis. A more realistic setting, which has first been defined in [JRS17] and is now used in the updated version of [Cho+22b], is to allow the adversary to query partial decryptions even *after* having seen the challenge ciphertext *and* to allow queries that involve the challenge ciphertext. Note that [JRS17] split the IND-CPA game into sequential phases, where the adversary first sends all messages to be encrypted at once and in a second phase sends all circuits to be evaluated and then partially decrypted, again at once. We strengthen the security notion by giving the adversary adaptive access to both the encryption and the partial decryption oracles simultaneously. Of course, to prohibit trivial attacks, the partial decryption oracle refuses to answer to queries which would directly leak which message has been encrypted when computing the challenge ciphertext. This flavour of security notion, while lacking simulation-based security, still offers a strong guarantee in the form of *input indistinguishability*: given partial decryptions for an evaluation $f(x_1, x_2)$, where x_1 is known to the adversary and x_2 is hidden, our security game implies that the adversary cannot distinguish whether the input x_2 was used, or some other input x_2' such that $f(x_1, x_2) = f(x_1, x_2')$. Similar notions have been used in secure multi-party computation [MPR06, CPP16].

To further motivate our definition, we highlight that allowing partial decryption queries that involve the challenge ciphertext is critical to achieving a meaningful notion of security. In a typical use-case, the goal of using ThFHE is to compute some function $f(x_1, \ldots, x_n)$, the result of which only reveals a small amount of information compared to the inputs x_i. However, in a security game it is always the challenge ciphertext that contains the hidden information, so disallowing this in partial decryption queries does not capture the desired goals. Indeed, consider the following ThFHE scheme that is obviously insecure in this setting: firstly, modify the evaluation algorithm to output not only an encryption of $f(x_1, \ldots, x_n)$, but also the encryption of x_1; secondly, modify the partial decryption algorithm to also output partial decryptions for x_1. Given a set of partial decryptions for $f(x_1, \ldots, x_n)$, the parties will also learn x_1 which is

exactly what we want to avoid. Going back to the definition of IND-CPA security for ThFHE, as the security game of [Cho+22b] only allows for partial decryption queries before seeing the challenge ciphertext, the above obviously insecure construction could actually be shown secure using their definition.

Defining OW-CPA *Security for* ThFHE *(Sect. 3).* As mentioned above, the Rényi divergence is hard to use in the context of decision-based security notions, such as IND-CPA. We give some intuition on why this is the case in the following. The probability preservation property of RD allows us to reason about the probability of a bad event happening in two different games. Roughly speaking, this says that if D_1, D_2 are distributions such that the Rényi divergence of D_1 from D_2 is at most δ, then for any event E, it holds that $\Pr[D_1(E)] \leq (\Pr[D_2(E)] \cdot \delta)^c$, for some constant c close to 1. If the event E occurs with negligible probability in game D_2, then we can get by with a polynomial-sized δ to argue the same holds in D_1. However, this is inherently hard to make use of in distinguishing games like IND-CPA, where probabilities of winning are close to $1/2$.

Instead of IND-CPA security, therefore we first aim for OW-CPA security, which is easier to prove with the Rényi divergence. When defining OW-CPA in the (fully homomorphic) threshold setting, the main changes are that the adversary also obtains t shares of the secret key and has access to a bounded number of partial decryption queries. In order to avoid trivial attacks, the partial decryption oracle refuses to answer to queries which would leak too much information on the challenge messages which the adversary tries to recover. As a measurement of *too much information* we use conditional min-entropy [Dod+08]. In other words, the oracle only answers to queries if the min-entropy of the challenge message conditioned on all the previously queried circuits and circuit evaluations is not much smaller then the original min-entropy of the challenge message.

Constructing Full-Threshold OW-CPA-*Secure* ThFHE *(Sect. 5).* To simplify the presentation in the introduction, we first describe our construction in the full-threshold setting and then explain how to get t-out-of-n threshold. As a starting point, we take any encryption scheme whose decryption function is nearly linear, as is the case for most LWE-based encryption schemes (including FHE). That is, for a given ciphertext ct on a message m with respect to a key pair $(\mathsf{sk}, \mathsf{pk})$, it holds that $\langle \mathsf{sk}, \mathsf{ct} \rangle = m + e_{\mathsf{ct}}$, where e_{ct} is what we earlier called decryption noise and depends on the ciphertext and the secret key.[4]

To achieve threshold decryption, we use standard additive secret sharing to split the secret key into $\mathsf{sk}_1, \ldots, \mathsf{sk}_n$ in a setup phase. By linearity, we could simply set the partial decryption shares as $\tilde{d}_i = \langle \mathsf{ct}, \mathsf{sk}_i \rangle$. However, after summing all shares together, the parties recover e_{ct}, which leaks information on sk. As in previous threshold solutions for lattice-based schemes, to compute their decryption share d_i every party now locally adds to $\tilde{d}_i$ a noise term e_i which is sampled from the noise flooding distribution $\mathcal{D}_{\mathsf{flood}}$. When summing those partial decryption shares together, the parties learn $m + e_{\mathsf{ct}} + \sum_{i=1}^{n} e_i$.

[4] Actually, it only reveals an encoding of m, which is easy to decode as long as parameters are set accordingly.

To prove the OW-CPA security of our construction, we modify the security experiment such that in a first step, the answers to the partial decryption queries no longer depend on the underlying secret key sk (reflected by e_{ct}), and in a second step the secret key shares are also independent of sk. In this case, OW-CPA security of the threshold scheme is implied by the OW-CPA security of the underlying standard encryption scheme. We simulate the partial decryption noise term $e_{\mathsf{ct}} + \sum_{i=1}^{n} e_i$ by sampling some independent noise $e' \leftarrow \mathcal{D}_{\mathsf{sim}}$. As long as the Rényi divergence between the two noise distributions is bounded by a constant, we can appeal to the probability preservation property, and the negligible probability of some PPT adversary guessing the message is preserved in both games. Note that previous works always chose $\mathcal{D}_{\mathsf{sim}} = \mathcal{D}_{\mathsf{flood}}$, but we later exploit in Sect. 6 that choosing a different $\mathcal{D}_{\mathsf{sim}}$ can lead to better parameters.

From Full-Threshold to t-Out-of-n Threshold (Sect. 5). When moving to the t-out-of-n setting, a natural choice is to use Shamir secret sharing. However, this leads to the problem that reconstruction is no longer addition, and instead requires multiplying the partial decryptions with Lagrange interpolation coefficients. These coefficients may be large, which in turn blows up the noise, breaking correctness. We offer two different solutions to this issue.

First, as in [Bon+18], we can use a special type of linear secret sharing scheme with binary coefficients, so that reconstruction is always a simple sum. Efficient threshold schemes with this property exist, for any n, t. We also consider a second method based on *pseudorandom secret sharing* [CDI05], which allows the parties to generate sharings of bounded, pseudorandom values without interaction. This uses replicated secret sharing, which is more expensive, but on the other hand, allows the partial decryptions to be converted into Shamir sharings before reconstruction. This leads to smaller partial decryptions, slightly better parameters and gives a form of robustness via Shamir error correction.

The high level idea is to encrypt a message m of δ bits, is to sample a random message x and to encrypt it using the OW-CPA-secure scheme. Then, the message bits are hidden by δ hard-core bits coming from a concatenation of δ Goldreich-Levin extractors. We use the notion of unpredictable entropy [HLR07] to give a bound on how many pseudorandom bits can be extracted from this construction. We say that a message x has unpredictability entropy k if for any PPT adversary $\mathcal{A}$ the probability of finding x given $\mathsf{Enc}(\mathsf{pk}, x)$ is at most 2^{-k}. We can then use existing results that show that a concatenation of δ Goldreich-Levin extractors can be used to extract $k - O(\log(1/\varepsilon))$ pseudorandom bits, where ε is the desired distinguishing advantage. Those pseudorandom bits then allow us to encrypt a message such that the ciphertexts of two given messages are computationally indistinguishable.

To prove this construction IND-CPA secure, we additionally need to assume circuit privacy of the underlying OW-CPA secure FHE scheme. Intuitively, this is necessary because the IND-CPA security definition says that an adversary should not be able to distinguish between the partial decryptions of a ciphertext encrypting $f(x_1, x_2)$ and those for a ciphertext encrypting $f(x_1, x_2')$ for some $x_2' \neq x_2$ where $f(x_1, x_2) = f(x_1, x_2')$. If, for instance, $x_2' = x_2 \oplus 1$, this is equiva-

lent to distinguishing between ciphertexts for $f(x_1, x_2)$ and $g(x_1, x_2)$, where the function g is defined as $g(x, y) = f(x, y \oplus 1)$. This can be seen as a circuit privacy problem, thus, intuitively, it seems that some form of circuit privacy is necessary to build IND-CPA-secure FHE.

Sample Parameters and Security Analysis (Sect. 6). We conclude our work by discussing how to choose concrete sample parameters for our threshold PKE scheme, when instantiating it with the lattice-based scheme Kyber [Sch+20].

As an example, to obtain 1-out-of-2 threshold decryption with a single query (e.g. for ephemeral key exchange), we can use the same parameters as Kyber1024 with a modulus increased only by a factor of 5, while supporting > 100 bits of classical hardness from our reduction. In a setting with up to 2^{32} queries, we need to use a 39-bit modulus and slightly larger module rank; this increases the ciphertext size by around 5x.

1.2 Related Work

Similarly to our work, [Cho+22b] used the Rényi divergence to obtain threshold FHE from LWE with a polynomial modulus-to-noise ratio. By arguing that the public sampleability property applies in their setting, they directly used the Rényi divergence to prove IND-CPA security. However, their definition of IND-CPA security is weaker than ours with respect to several aspects: First, they only allow for static access to the different queries, e.g., the adversary has to send the full list of circuits to the partial decryption oracle at once. In our case, those queries are adaptive, i.e., the adversary can adapt their next query to the partial decryption oracle based on the outcome of the previous query. Second, their adversary has to query challenge encryptions *before* querying partial decryptions. Moreover, their work focuses on a specific construction of ThFHE based on Torus-FHE, whereas our results are phrased generically for all encryption schemes with nearly linear decryption. Lastly, they focus on linear integer secret sharing schemes, whereas we additionally propose pseudorandom secret sharing and different ways of achieving robustness.

The Rényi divergence has seen widespread use in security proofs in lattice-based cryptography, since [Bai+18]. Replacing statistical noise flooding by Rényi noise flooding has led to a significant improvement in parameters for security reductions, for instance when proving the hardness of (structured) LWE with a binary secret [Bou+20], when designing multi-key FHE [DWF22], or more recently, in the context of lattice-based threshold signatures [ASY22]. The latter work of [ASY22] is quite similar to ours, since they also apply Rényi noise flooding to threshold FHE; however, they do not directly prove security of the threshold FHE scheme, and instead analyze the resulting threshold signature scheme directly (which is based on a search problem, so amenable to a Rényi divergence analysis). They additionally show the optimality of their noise flooding by providing an attack when a smaller noise flooding ratio is used. As the attack uses that their signature scheme is deterministic, it does not directly apply to our randomized encryption scheme. Previous works [Dev+21b, Nae+20] have already observed that OW-CPA allows to bypass the issues caused by the Rényi divergence. However, both are in the PKE setting, whereas our work focused

on the FHE setting. This required some care: it is not straightforward to define a OW-CPA notion in the fully-homomorphic setting and the standard transformation used in [Dev+21b, Nae+20] to lift one-way security to indistinguishability is not suited for the fully-homomorphic setting neither.

In an independent line of work, another noise flooding technique, called gentle noise flooding, has been studied in order to avoid the superpolynomial parameter blow-up [BD20a]. It was first used in theoretical hardness results on entropic (structured) LWE [BD20a, BD20b]. Later, a similar technique was used in [Cas+22] for improving parameters in additively homomorphic encryption with circuit privacy. The setting of [Cas+22] is quite different to ours, however, since with circuit privacy, the challenge is to deal with leakage on a plaintext rather than the secret key. This is handled via gentle noise flooding by applying a randomized encoding to the plaintext, so that leaking a constant fraction of its coordinates does not reveal anything about the plaintext. A similar technique does not seem to work in the threshold setting, with leakage on the secret key.

From a high level perspective, our notion of IND-CPA security has some similarities to the notion of IND-CPA^D security introduced in [LM21] in the context of approximate FHE. For instance, partial decryption queries in our setting correspond to decryption queries in their setting. Our security notion further matches with the game-based input-indistinguishability notion in the context of secure multi-party computation from [MPR06, CPP16], when realizing the latter with the help of ThFHE.

Another approach to build threshold key generation and decryption protocols is to use general multi-party computation tools like garbled circuits. This was done in [Kra+19] for a Ring-LWE based scheme. Their solution does not need any noise flooding or increased parameters of the underlying scheme, however, it relies on generic multi-party computation techniques like garbled circuits, and the partial decryption shares are generated using an expensive, interactive protocol rather than non-interactively as in our setting.

2 Preliminaries

For any positive integer q, we denote by $\mathbb{Z}_q$ the integers modulo q and for any positive integer n, we denote by $[n]$ the set $\{1, \dots, n\}$. Vectors are denoted in bold lowercase and matrices in bold capital letters. The identity matrix of order m is denoted by $\mathbf{I}_m$. The concatenation of two matrices $\mathbf{A}$ and $\mathbf{B}$ with the same number of rows is denoted by $[\mathbf{A}|\mathbf{B}]$. The abbreviation PPT stands for probabilistic polynomial-time. When we split a PPT adversary $\mathcal{A}$ in several sub algorithms $(\mathcal{A}_i)_i$, we implicitly assume that $\mathcal{A}_i$ outputs a state that is passed to the next $\mathcal{A}_{i+1}$. We call a function $\mathsf{negl}(\cdot)$ negligible in λ if $\mathsf{negl}(\lambda) = \lambda^{-\omega(1)}$, i.e., it decreases faster towards 0 than the inverse of any polynomial.

Throughout the paper we make use of the random oracle model (ROM), where we assume the existence of perfectly random functions, realized by oracles. For a random oracle $\mathsf{F}\colon \{0,1\}^n \to \{0,1\}^m$ it holds that $\Pr[\mathsf{F}(x) = y] = 2^{-m}$ and that $\Pr[\mathsf{F}(x) = \mathsf{F}(x') = y \colon x \neq x'] = \Pr[\mathsf{F}(x) = y] \cdot \Pr[\mathsf{F}(x') = y] = 2^{-2m}$. Hence, random oracles are per definition collision resistant. For $x, y \in \{0,1\}^n$ we denote by $x \oplus y$ the bit-wise XOR operator.

2.1 Probability and Entropy

For a finite set S, we denote its cardinality by $|S|$ and the uniform distribution over S by $U(S)$. The operation of sampling an element $x \in S$ according to a distribution D over S is denoted by $x \leftarrow D$, where the set S is implicit.

For standard deviation $\sigma > 0$ and mean $c \in \mathbb{R}$, we define the continuous Gaussian distribution $D_{\sigma,c} \colon \mathbb{R} \to (0,1]$ by $D_{\sigma,c}(x) = 1/(\sigma\sqrt{2\pi}) \cdot \exp(-(x-c)^2/(2\sigma^2))$. We also define the rounded Gaussian distribution over $\mathbb{Z}$, by rounding the result to the nearest integer, and denote this by $\lfloor D_{\sigma,c} \rceil$.

A random variable X over $\mathbb{R}$ is called τ-subgaussian for some $\tau > 0$ if for all s it holds $\mathbb{E}[\exp(sX)] \leq \exp(\tau^2 s^2/2)$. A τ-subgaussian random variable satisfies $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^2] \leq \tau^2$. We associate to X the width $\sigma = \sqrt{\mathbb{E}[X^2]}$. The continuous Gaussian distribution D_σ and its rounded version $\lfloor D_\sigma \rceil$ are σ-subgaussian. Further, the uniform distribution over $[-a,a] \cap \mathbb{Z}$ is a-subgaussian.

The statistical distance between two probability distributions X and Y, denoted by $\mathsf{sdist}(X,Y)$, is defined as $\max_T |\Pr[T(X)=1] - \Pr[T(Y)=1]|$, where T is any test function. The computational distance with respect to size s circuits, denoted by $\mathsf{cdist}_s(X,Y)$, limits T to be any circuit of size s. For any event E, the probability preservation property of sdist (resp. cdist_s) states that $X(E) \leq Y(E) + \mathsf{sdist}(X,Y)$ (resp. $X(E) \leq Y(E) + \mathsf{cdist}_s(X,Y)$).

The notion of unpredictable entropy has been introduced and studied in [HLR07] in the context of conditional computational entropy.

Definition 1 (Unpredictable Entropy). *For a distribution (X,Z), we say that X has* unpredictable entropy *at least k conditioned on Z, if there exists a collection of distributions Y_Z (giving rise to a joint distribution (Y,Z)) such that $\mathsf{cdist}_s((X,Z),(Y,Z)) \leq \varepsilon$, and for all circuits C of size s,*

$$\Pr[C(Z) = Y] \leq 2^{-k}.$$

We write $H^{\mathsf{unp}}_{\varepsilon,s}(X|Z) \geq k$.

Definition 2 (Concatenated Goldreich-Levin Extractor). *Fix $n, \delta \in \mathbb{N}$. We define the* concatenated Goldreich-Levin extractor *$\mathcal{E} \colon \{0,1\}^n \times (\{0,1\}^n)^\delta \to \{0,1\}^\delta \times (\{0,1\}^n)^\delta$ as*

$$\mathcal{E}(x, s_1, \ldots, s_\delta) := (\langle x, s_1 \rangle \bmod 2, \ldots, \langle x, s_\delta \rangle \bmod 2, s_1, \ldots, s_\delta).$$

By analyzing a *reconstruction property* of the Goldreich-Levin extractor and applying a result of [HLR07], we get the following concrete bound on the number of pseudorandom bits that can be extracted.

Lemma 1 ([HLR07], Lemma 6). *Let X be a distribution with unpredictable entropy $H^{\mathsf{unp}}_{\varepsilon,s}(X|Z) \geq k$ and let $\mathcal{E}$ be the concatenated Goldreich-Levin extractor for some $n, \delta \in \mathbb{N}$. If $k = \delta + \log_2 2n + 3\log_2 1/\varepsilon$, then $\mathcal{E}$ extracts δ pseudorandom bits, i.e.,*

$$\mathsf{cdist}_{s'}\left((Z, \mathcal{E}(X, U(\{0,1\}^{n\delta}))), (Z, U(\{0,1\}^\delta \times \{0,1\}^{n\delta}))\right) \leq 5\delta\varepsilon,$$

where $s' = O(sn^{-3}\varepsilon^4)$.

Let $\mathbf{x}$ follow a distribution on a set X, and $\mathbf{z}$ follow a possibly correlated distribution on a set Z. The *average conditional min-entropy* [Dod+08] of $\mathbf{x}$ given $\mathbf{z}$ is defined by

$$\widetilde{H}_\infty(\mathbf{x}|\mathbf{z}) = -\log_2\left(E_{\mathbf{z}'}\left[\max_{\mathbf{x}'\in X}\Pr[\mathbf{x}=\mathbf{x}'|\mathbf{z}=\mathbf{z}']\right]\right).$$

Lemma 2 ([Dod+08], Lem. 2.2). *Let $\mathbf{x}, \mathbf{y}, \mathbf{z}$ be three random variables, where $\mathbf{z}$ takes at most 2^λ values. Then*

$$\widetilde{H}_\infty(\mathbf{x}|\mathbf{y}, \mathbf{z}) \geq \widetilde{H}_\infty(\mathbf{x}|\mathbf{y}) - \lambda.$$

The Rényi divergence (RD) defines an alternative measure of distribution closeness. We follow [Bai+18] and use a definition of the RD which is the exponential of the classical definition. We restrict the order a to be in $(1, \infty)$.

Definition 3 (Rényi Divergence). *Let P and Q be two discrete probability distributions such that* $\mathrm{Supp}(P) \subseteq \mathrm{Supp}(Q)$. *For $a \in (1, \infty)$ the* Rényi divergence of order a *is defined by*

$$\mathrm{RD}_a(P, Q) = \left(\sum_{x\in\mathrm{Supp}(P)} \frac{P(x)^a}{Q(x)^{a-1}}\right)^{\frac{1}{a-1}}.$$

The definitions are extended in the natural way to continuous distributions. We recall some useful properties of the RD. The first two were proven in [EH14] and the last one was proven in [Ros20, Prop. 2].

Lemma 3. *Let P, Q be two discrete probability distributions with* $\mathrm{Supp}(P) \subseteq \mathrm{Supp}(Q)$. *For $a \in (1, \infty)$, it yields:*

Data Processing Inequality: $\mathrm{RD}_a(g(P)\|g(Q)) \leq \mathrm{RD}_a(P\|Q)$ *for any function g, where $g(P)$ (resp. $g(Q)$) denotes the distribution of $g(y)$ induced by sampling $y \leftarrow P$ (resp. $y \leftarrow Q$).*

Probability Preservation: *Let $E \subset \mathrm{Supp}(Q)$ be an event, then for $a \in (1, \infty)$*

$$Q(E) \cdot \mathrm{RD}_a(P\|Q) \geq P(E)^{\frac{a}{a-1}}.$$

Multiplicativity: *Let P, Q be two probability distributions of a pair of random variables (Y_1, Y_2). For $i \in \{1, 2\}$, let P_i (resp. Q_i) denote the marginal distribution of Y_i under P (resp. Q), and let $P_{2|1}(\cdot|y_1)$ (resp. $Q_{2|1}(\cdot|y_1)$) denote the conditional distribution of Y_2 given that $Y_1 = y_1$. Then for $a \in (1, \infty)$*

$$\mathrm{RD}_a(P\|Q) \leq \mathrm{RD}_a(P_1\|Q_1) \cdot \max_{y_1\in Y_1} \mathrm{RD}_a(P_{2|1}(\cdot|y_1)\|Q_{2|1}(\cdot|y_1)).$$

The Rényi divergence of two shifted Gaussians is given below. This also allows us to bound the RD of rounded Gaussians by the data processing inequality.

Lemma 4 ([GAL13]). *Let σ be a positive real number and $c \in \mathbb{Z}$. Then for $a \in (1, \infty)$ it yields*

$$\mathrm{RD}_a(D_{\sigma,c} \| D_\sigma) = \exp\left(\frac{ac^2}{2\sigma^2}\right).$$

We provide the proof of the following lemma in the full version [BS23, Sec. 2].

Lemma 5. *Let D_1, D_2 be two probability distributions over $\mathbb{Z}$ and $e_1, \ldots, e_N$ be (possibly dependent) random variables over $\mathbb{Z} \cap [-B, B]$ for some $B \in \mathbb{Z}$, for which there exist $a \in (1, \infty)$ and $\rho \geq 1$ such that for all β with $|\beta| \leq B$, it holds that $\mathrm{Supp}(D_1 + \beta) \subseteq \mathrm{Supp}(D_2)$, and furthermore, $\mathrm{RD}_a(D_1 + \beta \| D_2) \leq \rho$. Then,*

$$\mathrm{RD}_a((D_1 + e_N, \ldots, D_1 + e_1) \| D_2^N) \leq \rho^N.$$

2.2 Linear Secret Sharing

We use linear secret sharing schemes (LSSS) for monotone access structures with a special $\{0,1\}$-reconstruction property, as follows.

Definition 4 (Monotone Access Structure). *Let $\mathcal{P} = \{P_1, \ldots, P_n\}$ be a set of parties and $2^{\mathcal{P}}$ its power set. A* monotone access structure *is a collection of sets $\mathbb{A} \subset 2^{\mathcal{P}}$, such that for any $S \in \mathbb{A}$, if $T \supset S$ then $T \in \mathbb{A}$. We say that $\mathbb{A}$ is* efficient *if membership of $\mathbb{A}$ can be verified in time $\mathsf{poly}(\lambda)$, where $\mathbb{A}$ is viewed as a function of λ.*

In this work, we only consider efficient access structures. To ease notation, we identify a party P_i with its index i, viewing each set $S \in \mathbb{A}$ as a subset of $[n]$. For any $S \subset [n]$ and vector $\mathbf{v} = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$, we let $\mathbf{v}|_S$ denote the vector of shares restricted to $\mathbf{v}_i$ for indices $i \in S$.

Definition 5 (Linear Secret Sharing Scheme). *Let q, L, n be positive integers and $\mathbb{A}$ a monotone access structure. A linear secret sharing scheme* LSSS *for $\mathbb{A}$ is defined by a randomized algorithm $\mathsf{Share} : \mathbb{Z}_q \to (\mathbb{Z}_q^L)^n$ and a family of deterministic algorithms $\mathsf{Rec}_S : (\mathbb{Z}_q^L)^{|S|} \to \mathbb{Z}_q$, for $S \subseteq [n]$, which satisfy:*

Privacy: *For any set $S \notin \mathbb{A}$, any $x, x' \in \mathbb{Z}_q$ and $\mathbf{v} \in \mathbb{Z}_q^{L|S|}$, it holds that $\Pr[\mathsf{Share}(x)|_S = \mathbf{v}] = \Pr[\mathsf{Share}(x')|_S = \mathbf{v}]$.*

Reconstruction: *For any set $S \in \mathbb{A}$, any $x \in \mathbb{Z}_q$ and $\mathbf{v} = \mathsf{Share}(x)$, the reconstruction algorithm outputs $\mathsf{Rec}_S(\mathbf{v}|_S) = x$.*

Linearity: *For any $\alpha, \beta \in \mathbb{Z}_q$, any set S with $|S| > t$ and any share vectors $\mathbf{u}, \mathbf{v}$, it holds that $\mathsf{Rec}_S(\alpha\mathbf{u}|_S + \beta\mathbf{v}|_S) = \alpha\mathsf{Rec}_S(\mathbf{u}|_S) + \beta\mathsf{Rec}(\mathbf{v}|_S)$.*

When the set of shares is $S = [n]$, we write Rec instead of $\mathsf{Rec}_{[n]}$.

We need the following notion of valid and invalid share sets [Bon+18].

Definition 6. *Let $x \in \mathbb{Z}_q$, $(\mathbf{v}_1, \ldots, \mathbf{v}_n) = \mathsf{Share}(x)$, and write $\mathbf{v}_i = (\mathbf{v}_{i,1}, \ldots, \mathbf{v}_{i,L})$. A set of pairs of indices $T \subseteq [n] \times [L]$ is an* invalid set of share elements *if the corresponding shares $(\mathbf{v}_{i,j})_{(i,j)\in T}$ reveal no information about x. Otherwise, we say that T is a* valid set of share elements. *We additionally say:*

- $T \subseteq [n] \times [L]$ *is a* maximal invalid set of share elements *if it is invalid, but for any* $(i,j) \in [n] \times [L] \setminus T$*, the set* $T \cup \{(i,j)\}$ *is a valid set of share elements.*
- $T \subseteq [n] \times [L]$ *is a* minimal valid set of share elements *if it is valid, but for any* $T' \subsetneq T$*, the set* T' *is an invalid set of share elements.*

Note that in any LSSS, a valid set as defined above always allows reconstruction of the secret x. This is because an LSSS can equivalently be defined by a matrix M, such that each share element $\mathbf{v}_{i,j}$ is computed as the inner product of some row of M and $(x, r_1, \ldots, r_{n-1})$, where r is the randomness used in Share. Reconstruction is possible for a given set of share elements iff the corresponding set of rows of M span the target vector $(1, 0, \ldots, 0)$. This definition implies that any set of rows is either invalid—and reveals nothing about x—or valid, and allows full reconstruction. For further details, see e.g. [Bei96, Chapter 4].

Our main construction requires that the reconstruction function Rec_S takes a $0/1$ combination of its inputs. In the following, we require this to hold not only for any set of shares corresponding to a valid set of parties in $\mathbb{A}$, but for any valid set of share elements. This property is equivalent to the notion of a derived $\{0,1\}$-LSSS, used in [JRS17].[5]

Definition 7 (Strong $\{0,1\}$-Reconstruction). *We say that a* LSSS *has* strong {0,1}-reconstruction *if for any secret* x *and* $(\mathbf{v}_1, \ldots, \mathbf{v}_n) = \mathsf{Share}(x)$*, for any valid set of share elements* $T \subseteq [n] \times [L]$*, there exists a subset* $T' \subseteq T$ *such that* $\sum_{(i,j) \in T'} \mathbf{v}_{i,j} = x$*, where* $\mathbf{v}_i = (\mathbf{v}_{i,1}, \ldots, \mathbf{v}_{i,L})$*.*

Sharing Values in R_q. In our constructions, we share $\mathbf{x} \in R_q^r$, where $R_q = \mathbb{Z}_q[X]/f(X)$, instead of just in $\mathbb{Z}_q$. We do this coefficient-wise, by separately sharing each coefficient of the r polynomials in $\mathbf{x}$. Each party's share then lies in $(R_q^r)^L$, and the parties can perform R_q-linear operations on these shares.

Example Linear Secret Sharing Schemes. In Table 1, we detail a few example secret sharing schemes we consider. The schemes are for t-out-of-n access structures, where any $t+1$ parties can reconstruct, and they all have strong $\{0,1\}$-reconstruction. In the table, we show two quantities $\tau_{\mathsf{max}}, \tau_{\mathsf{min}}$, which are relevant for choosing parameters in our constructions of Sect. 5 and we will refer to later. By τ_{max} we denote the size of the smallest maximal invalid set of share elements, while τ_{min} is the size of the largest minimal valid set of share elements.

[5] [Bon+18] only assumed a weaker property for their threshold FHE construction. However, this is a mistake introduced when merging the two works [JRS17] and [Bon+17] (and has been confirmed by the authors of [JRS17]).

Table 1. Example t-out-of-n linear secret sharing schemes with strong $\{0,1\}$-reconstruction. Details for the last row are omitted, due to their complexity.

Scheme	Sharing method	P_i's share	L	τ_{max}	τ_{min}
Additive	$x = \sum_{i=1}^{n} x_i$	x_i	1	$n-1$	n
Replicated	$x = \sum_{A,\lvert A\rvert=t} x_A$	$\{x_A\}_{i\notin A}$	$\binom{n-1}{t}$	$(n-t)(\binom{n}{t}-1)$	$\binom{n}{t}$
Naive	$x = \sum_{i\in A} x_{A,i}$, $\lvert A\rvert = t+1$	$\{x_{A,i}\}_{i\in A}$	$\binom{n-1}{t}$	$t\binom{n}{t+1}$	$t+1$
Monotone Boolean formula for threshold fn. [Val84]			$O(n^{4.3})$	$O(n^{5.3})$	$O(n^{5.3})$

2.3 Learning with Errors

In the following, we recall the definitions of the decision (module) LWE problem [Reg05, LS15], formulated with a bounded uniform secret and noise. Let $R_q = \mathbb{Z}_q[X]/f(X)$ for some irreducible $f(X)$ of degree d. Further, we define $S_\beta = \{a \in R\colon \|a\|_\infty \leq \beta\}$ with $\beta \in \mathbb{N}$.

Definition 8 (M-LWE). *Let $m, r, \beta, q \in \mathbb{N}$. The* Module Learning With Errors *problem* M-LWE$_{q,m,r,\beta}$ *is defined as follows. Given $\mathbf{A} \leftarrow U(R_q^{m\times r})$ and $\mathbf{t} \in R_q^m$. Decide whether $\mathbf{t} \leftarrow U(R_q^m)$ or if $\mathbf{t} = [\mathbf{A}|\mathbf{I}_m]\cdot\mathbf{s}$, where $\mathbf{s} \leftarrow U(S_\beta^{m+r})$.*

The special case of $d = 1$, where the ring R is isomorphic to $\mathbb{Z}$, is simply denoted LWE (and is historically the one that has been introduced first).

3 Threshold Fully Homomorphic Encryption

3.1 Syntax and Basic Properties of Threshold FHE/PKE

We first recall the syntax of a fully homomorphic threshold public key encryption scheme. We implicitly assume that after Setup, all algorithms are given the public parameters as input. We omit the partial verification algorithm used in previous works (e.g., [BBH06]), which was only used to model stronger notions of robustness that also capture CCA attacks.

Definition 9 (ThFHE). *A* fully homomorphic threshold public key encryption *scheme* (ThFHE) *for a message space $\mathcal{M}$ and circuits of depth κ is a tuple of* PPT *algorithms* $\mathsf{ThFHE} = (\mathsf{Setup}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{PartDec}, \mathsf{Combine})$ *defined as follows:*

$\mathsf{Setup}(1^\lambda, 1^\kappa, n, t) \to (\mathsf{pp}, \mathsf{pk}, \mathsf{sk}_1, \ldots, \mathsf{sk}_n)$*: On input the security parameter λ, a bound on the circuit depth κ, the number of parties n and a threshold value $t \in \{1, \ldots, n-1\}$, the setup algorithm outputs the public parameters* pp*, a public key* pk *and a set of secret key shares* $\mathsf{sk}_1, \ldots, \mathsf{sk}_n$*.*

$\mathsf{Enc}(\mathsf{pk}, m) \to \mathsf{ct}$*: On input the public key* pk *and a message $m \in \mathcal{M}$, the encryption algorithm outputs a ciphertext* ct*.*

$\mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct}_1, \ldots, \mathsf{ct}_k) \to \mathsf{ct}$*: On input the public key* pk*, a circuit* $C\colon \mathcal{M}^k \to \mathcal{M}$ *of depth at most* κ *and a set of ciphertexts* $\mathsf{ct}_1, \ldots, \mathsf{ct}_k$*, the evaluation algorithm outputs a ciphertext* ct*.*

$\mathsf{PartDec}(\mathsf{sk}_i, \mathsf{ct}) \to d_i$*: On input a key share* sk_i *for some* $i \in [n]$ *and a ciphertext* ct*, the partial decryption algorithm outputs a partial decryption share* d_i*.*

$\mathsf{Combine}(\{d_i\}_{i\in S}, \mathsf{ct}) \to m'$*: On input a set of decryption shares* $\{d_i\}_{i\in S}$ *and a ciphertext* ct*, where* $S \subset [n]$ *is of size at least* $t+1$*, the combining algorithm outputs a message* $m' \in \mathcal{M} \cup \{\perp\}$*.*

The above can be seen as a generalization encompassing non-threshold and threshold PKE and FHE.

Definition 10 (ThPKE). *A* threshold public key encryption *scheme* (ThPKE) *for a message space* $\mathcal{M}$ *is a* ThFHE *scheme, where* $k = 1$ *and the only allowed circuit* $C\colon \mathcal{M} \to \mathcal{M}$ *is the identity. In this case, we drop the trivial evaluation algorithm* Eval *and the parameter* κ *in the scheme's specifications.*

Definition 11 (FHE). *A* fully homomorphic public key encryption *scheme* (FHE) *for a message space* $\mathcal{M}$ *is a* ThFHE *scheme, where* $n = 1$*. In this case, we drop the parameters* n *and* t *in the scheme's specifications. To simplify notations, we merge* PartDec *and* Combine *into one single algorithm that we denote* Dec*. Hence, the algorithm* Dec *takes* sk *and* ct *as input and outputs* $m' \in \{\mathcal{M} \cup \{\perp\}\}$*.*

We require compactness and correctness, whose definitions we recall in the full version [BS23, App. A]. There, we also define two notions of robustness [BS23, Sec. 3.2].

In Sect. 4, we also need FHE schemes which are circuit private, defined below. This can be instantiated under LWE with a polynomial modulus [DS16].

Definition 12 (Circuit Privacy). *Let* $s, \varepsilon > 0$*. A* ThFHE *scheme with message space* $\mathcal{M}$ *and maximal circuit depth* κ *fulfills* (s, ε)-circuit privacy *if for every circuit* C *of depth at most* κ *it yields*

$$\mathsf{cdist}_s\left(((\mathsf{sk}_i)_{i\in n}, \mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct}_1, \ldots, \mathsf{ct}_k)), ((\mathsf{sk}_i)_{i\in[n]}, \mathsf{Enc}(\mathsf{pk}, C(m_1, \ldots, m_k)))\right) \leq \varepsilon,$$

where $m_i \in \mathcal{M}$ *and* $\mathsf{ct}_i \leftarrow \mathsf{Enc}(\mathsf{pk}, m_i)$ *for all* $i \in [k]$ *and for honestly generated keys* $(\mathsf{pk}, \mathsf{sk}_1, \ldots, \mathsf{sk}_n)$*.*

3.2 One-Wayness

We now present our definition of OW-CPA security for ThFHE schemes.

The high level idea of the security game is the following. At the beginning, the adversary decides on the parties they want to corrupt and receives the corresponding secret key shares. We call this the static corruption setting. Then the adversary has access to three different oracles. The first, OEnc, allows them to obtain honestly generated, fresh ciphertexts on messages of their choice. Through

the second oracle, $\mathsf{OChallEnc}$, the adversary obtains encryptions of unknown, randomly chosen messages, which we call the challenge messages and challenge ciphertexts. Finally, they can query up to ℓ times the last oracle, $\mathsf{OPartDec}$, by inputting a circuit and a list of indices referring to previous encryption *and* challenge encryption queries, and receiving the corresponding partial decryption shares of all parties (after the evaluation algorithm has been applied). However, the partial decryption oracle aborts if for one of the challenge messages the conditional min-entropy has decreased more than an allowed amount ν, after having learned the circuit evaluation. Note that we do not condition the information-theoretical min-entropy on $\mathtt{ChallCT}$ as it uniquely defines $\mathtt{ChallM}$. Implicitly, we assume that the entropy condition can be efficiently verified for the circuits input to $\mathsf{OPartDec}$. One way to practically implement this, is to ask the adversary to input an algorithm which verifies the entropy condition when querying the oracle. We stress that in the transformation of Sect. 4 we only query circuits for which the entropy condition can be checked efficiently. To highlight the query bound ℓ and the entropy loss bound ν, we write (ℓ, ν)-$\mathsf{OW\text{-}CPA}$.

Definition 13 ((ℓ, ν)-$\mathsf{OW\text{-}CPA}$ for ThFHE). *We call a* ThFHE *scheme* (ℓ, ν)-$\mathsf{OW\text{-}CPA}$ *secure for the security parameter* λ*, the circuit depth bound* κ*, the threshold parameters* n, t*, the query bound* ℓ *and the entropy bound* ν*, if for all* PPT *adversaries* $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$

$$\mathsf{Adv}^{(\ell,\nu)\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{ThFHE}}(\mathcal{A}) := \Pr[\mathsf{Expt}^{(\ell,\nu)\text{-}\mathsf{OW\text{-}CPA}}_{\mathcal{A},\mathsf{ThFHE}}(1^\lambda, 1^\kappa, n, t) = 1] = \mathsf{negl}(\lambda)\,,$$

where $\mathsf{Expt}^{(\ell,\nu)\text{-}\mathsf{OW\text{-}CPA}}_{\mathcal{A},\mathsf{ThFHE}}$ *is the experiment in Fig. 1 with* $\mathsf{ctr} = 0$*,* $\mathsf{idx} = 0$ *and* $\mathtt{L} = \emptyset$ *at the beginning.*

3.3 Indistinguishability

In the following, we present our definition of $\mathsf{IND\text{-}CPA}$ security for ThFHE.

As for the $\mathsf{OW\text{-}CPA}$ security, we allow for static corruptions and access to three different oracles. The first, OEnc, is the same as in the $\mathsf{OW\text{-}CPA}$ game. To the second oracle, $\mathsf{OChallEnc}$, the adversary inputs two messages and obtains the encryption of one of it. Finally, they can again query up to ℓ times $\mathsf{OPartDec}$, by inputting a circuit and a list of indices and receiving the corresponding partial decryption shares of all parties. This time, the partial decryption oracle aborts if the circuit evaluates to different values on the corresponding input messages to the $\mathsf{OChallEnc}$ oracle. To highlight the query bound ℓ, we write ℓ-$\mathsf{IND\text{-}CPA}$.

By allowing adaptive access to all three oracles, our definition can be seen as a strengthening of Definition 14 in [JRS17].

Definition 14 (ℓ-$\mathsf{IND\text{-}CPA}$ for ThFHE). *We call a* ThFHE *scheme* ℓ-$\mathsf{IND\text{-}CPA}$ *secure for the security parameter* λ*, the circuit depth bound* κ*, the threshold parameters* n, t *and the query bound* ℓ*, if for all* PPT *adversaries* $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$

$$\mathsf{Adv}^{\ell\text{-}\mathsf{IND\text{-}CPA}}_{\mathsf{ThFHE}}(\mathcal{A}) := \left|\Pr[\mathsf{Expt}^{\ell\text{-}\mathsf{IND\text{-}CPA}}_{\mathcal{A},\mathsf{ThFHE}}(1^\lambda, 1^\kappa, n, t) = 1] - \frac{1}{2}\right| = \mathsf{negl}(\lambda)\,,$$

$\mathsf{Expt}^{(\ell,\nu)\text{-OW-CPA}}_{\mathcal{A},\mathsf{ThFHE}}(1^\lambda, 1^\kappa, n, t)$

$(\mathsf{pp}, \mathsf{pk}, \mathsf{sk}_1, ..., \mathsf{sk}_n) \leftarrow \mathsf{Setup}(1^\lambda, 1^\kappa, n, t)$
$S \leftarrow \mathcal{A}_1(\mathsf{pp}, \mathsf{pk}) \colon S \subset [n] \wedge |S| \leq t$
$(m', j) \leftarrow \mathcal{A}_2^{\mathsf{OEnc},\mathsf{OChallEnc},\mathsf{OPartDec}}(\mathsf{pk}, \{\mathsf{sk}_i\}_{i\in S})$
$(b_j, m_j, \mathsf{ct}_j) := \mathtt{L}[j]$
return $m_j = m' \wedge b_j = 1$

$\mathsf{OEnc}(m)$

if $m \notin \mathcal{M}$ **then return** $\perp$
$\mathsf{idx} = \mathsf{idx} + 1$
$\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, m)$
$\mathtt{CT} = \mathtt{CT} \cup \{\mathsf{ct}\}$
$\mathtt{L}[\mathsf{idx}] := \{(0, m, \mathsf{ct})\}$
return ct

$\mathsf{OChallEnc}()$

$\mathsf{idx} = \mathsf{idx} + 1$
$m \leftarrow \mathcal{M}$
$\mathtt{ChallM} = \mathtt{ChallM} \cup \{m\}$
$\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, m)$
$\mathtt{ChallCT} = \mathtt{ChallCT} \cup \{\mathsf{ct}\}$
$\mathtt{L}[\mathsf{idx}] := \{(1, m, \mathsf{ct})\}$
return ct

$\mathsf{OPartDec}(C, \iota_1, \ldots, \iota_k)$

$\mathsf{ctr} = \mathsf{ctr} + 1$
if $\mathsf{ctr} > \ell$ **then return** $\perp$
if $\exists j \in [k] \colon \iota_j > |\mathtt{L}|$ **then return** $\perp$
if $\mathsf{depth}(C) > \kappa$ **then return** $\perp$
$(b_j, m_j, \mathsf{ct}_j) := \mathtt{L}[\iota_j], \quad j \in [k]$
$\mathsf{ct} \leftarrow \mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct}_1, \ldots, \mathsf{ct}_k)$
$d_i \leftarrow \mathsf{PartDec}(\mathsf{sk}_i, \mathsf{ct}), \quad i \in [n]$
$\mathbf{d} = (d_i)_{i\in[n]}$
for $m \in \mathtt{ChallM}$
  if $\widetilde{H}_\infty(m|\mathtt{E} \cup \{(C, C(m_1, \ldots, m_k))\}) < \widetilde{H}_\infty(m) - \nu$ **then**
    return $\perp$
$\mathtt{E} = \mathtt{E} \cup \{(C, C(m_1, \ldots, m_k))\}$
$\mathtt{PartD} = \mathtt{PartD} \cup \{\mathbf{d}\}$
return $\mathbf{d}$

Fig. 1. Experiments for (ℓ, ν)-OW-CPA security of ThFHE schemes.

where $\mathsf{Expt}^{\ell\text{-IND-CPA}}_{\mathcal{A},\mathsf{ThFHE}}$ *is the experiment in Fig. 2 with* $\mathsf{ctr} = 0$ *and* $L = \emptyset$ *at the beginning.*

$\mathsf{Expt}^{\ell\text{-IND-CPA}}_{\mathcal{A},\mathsf{ThFHE}}(1^\lambda, 1^\kappa, n, t)$

1 : $(\mathsf{pp}, \mathsf{pk}, \mathsf{sk}_1, ..., \mathsf{sk}_n) \leftarrow \mathsf{Setup}(1^\lambda, 1^\kappa, n, t)$
2 : $S \leftarrow \mathcal{A}_1(\mathsf{pp}, \mathsf{pk})\colon S \subset [n] \wedge |S| \leq t$
3 : $b \leftarrow \{0,1\}$
4 : $b' \leftarrow \mathcal{A}_2^{\mathsf{OEnc},\mathsf{OChallEnc},\mathsf{OPartDec}}(\mathsf{pk}, \{\mathsf{sk}_i\}_{i\in S})$
5 : **return** $b = b'$

$\mathsf{OEnc}(m)$

1 : **if** $m \notin \mathcal{M}$ **then return** $\perp$
2 : $\mathsf{idx} = \mathsf{idx} + 1$
3 : $\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, m)$
4 : $\mathtt{L}[\mathsf{idx}] := \{(m, m, \mathsf{ct})\}$
5 : **return** ct

$\mathsf{OChallEnc}(m^{(0)}, m^{(1)})$

1 : **if** $(m^{(0)}, m^{(1)}) \notin \mathcal{M} \times \mathcal{M}$ **then return** $\perp$
2 : $\mathsf{idx} = \mathsf{idx} + 1$
3 : $\mathsf{ct}_b \leftarrow \mathsf{Enc}(\mathsf{pk}, m^{(b)})$
4 : $\mathtt{L}[\mathsf{idx}] := \{(m^{(0)}, m^{(1)}, \mathsf{ct}_b)\}$
5 : **return** ct_b

$\mathsf{OPartDec}(C, \iota_1, \ldots, \iota_k)$

1 : $\mathsf{ctr} = \mathsf{ctr} + 1$
2 : **if** $\mathsf{ctr} > \ell$ **then return** $\perp$
3 : **if** $\exists j \in [k]\colon \iota_j > |\mathtt{L}|$ **then return** $\perp$
4 : **if** $\mathsf{depth}(C) > \kappa$ **then return** $\perp$
5 : $(m_j^{(0)}, m_j^{(1)}, \mathsf{ct}_j) := \mathtt{L}[\iota_j], \quad j \in [k]$
6 : **if** $C(m_1^{(0)}, \ldots, m_k^{(0)}) \neq C(m_1^{(1)}, \ldots, m_k^{(1)})$ **then return** $\perp$
7 : $\mathsf{ct} \leftarrow \mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct}_1, \ldots, \mathsf{ct}_k)$
8 : $d_i \leftarrow \mathsf{PartDec}(\mathsf{sk}_i, \mathsf{ct}), \quad i \in [n]$
9 : **return** $(d_i)_{i\in[n]}$

Fig. 2. Experiments for ℓ-IND-CPA security of ThFHE schemes.

4 From One-Wayness to Indistinguishability

In the following, we describe a generic way of transforming a OW-CPA secure ThFHE scheme into an IND-CPA secure one in the standard model, via hardcore bits.

The Construction. The transformation is parameterized by $\delta, \gamma \in \mathbb{N}$. Given $\mathsf{ThFHE} = (\mathsf{Setup}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{PartDec}, \mathsf{Combine})$ with message space $\mathcal{M} = \{0,1\}^\gamma$ being OW-CPA secure, we define $\mathsf{ThFHE}' = (\mathsf{Setup}', \mathsf{Enc}', \mathsf{Eval}', \mathsf{PartDec}', \mathsf{Combine}')$ with message space $\mathcal{M}' = \{0,1\}^\delta$, which fulfills IND-CPA security, as follows.

Setup': On input $(1^\lambda, 1^\kappa, n, t)$, it outputs $(\mathsf{pp}, \mathsf{pk}, \mathsf{sk}_1, \ldots, \mathsf{sk}_n) \leftarrow \mathsf{Setup}(1^\lambda, 1^\kappa, n, t)$.

Enc': On input (pk, m) with $m = (m_j)_{j\in[\delta]} \in \mathcal{M}'$, it samples $x \leftarrow U(\mathcal{M})$ and computes $c_0 \leftarrow \mathsf{Enc}(\mathsf{pk}, x)$. For $j \in [\delta]$, it samples $s_j \leftarrow U(\mathcal{M})$ and computes $c_j = \langle x, s_j\rangle + m_j \bmod 2$. It outputs $\mathsf{ct} = (c_0, s_1, \ldots, s_\delta, c_1, \ldots, c_\delta)$.

Eval': On input $I := (\mathsf{pk}, C, \mathsf{ct}_1, \ldots, \mathsf{ct}_k)$, where $\mathsf{ct}_i = (c_{i0}, s_{i1}, \ldots, s_{i\delta}, c_{i1}, \ldots, c_{i\delta})$ such that $c_{i0} \leftarrow \mathsf{Enc}(\mathsf{pk}, x_i)$ for $i \in [k]$ and $C\colon (\mathcal{M}')^k \to \mathcal{M}'$, it first defines a circuit $\widetilde{C}\colon (\mathcal{M})^k \to \mathcal{M}$ as follows:

- $\widetilde{C}$ takes as input $(x_1, \ldots, x_k)$ and has the information I hard-coded
- It computes $m_{ij} = c_{ij} + \langle x_i, s_{ij}\rangle \bmod 2$, for $j \in [\delta]$ and $i \in [k]$
- It outputs $C(m_1, \ldots, m_k)$, where $m_i = (m_{ij})_{j\in[\delta]}$

It then outputs $\mathsf{ct}' = \mathsf{Eval}(\mathsf{pk}, \widetilde{C}, c_{10}, \ldots, c_{k0})$.

$\mathsf{PartDec}'$: On input $(\mathsf{sk}_i, \mathsf{ct}')$, it outputs $d_i = \mathsf{PartDec}(\mathsf{sk}_i, \mathsf{ct}')$.

$\mathsf{Combine}'$: On input $(\{d_i\}_{i\in S}, \mathsf{ct}')$, it outputs $m = \mathsf{Combine}(\{d_i\}_{i\in S}, \mathsf{ct}')$.

Ciphertext Expansion. The ratio between the bit size of the plaintext and the ciphertext is give by

$$\frac{|\mathsf{ct}|}{|m|} = \frac{|c_0| + \delta(\gamma+1)}{\delta},$$

where c_0 is the OW-CPA ciphertext encrypting γ bits coming from ThFHE. We can see that with larger δ the ciphertext expansion gets better.

We prove compactness and decryption correctness in the full version [BS23, App. B].

Remark 1. One way to reduce the size of the ciphertext to $|c_0|+\gamma+\delta$ (and hence to improve the ciphertext expansion) is to replace the δ random seeds $s_1, \ldots, s_\delta$ by one single seed and a random oracle F. More precisely, one could define $s_j := \mathsf{F}(r, j)$ for a random seed $r \leftarrow U(\mathcal{M})$ and $j \in [\delta]$. As a result, the transformation wouldn't be in the standard, but in the random oracle model. As the random oracle is only used to derive the seeds, not when masking the message, this transformation still applies to the threshold FHE setting.

Theorem 1 (Security). *Fix $\ell, \delta, \gamma, \lambda, s \in \mathbb{N}$ and $\varepsilon > 0$. Let q_c denote the number of provided challenge ciphertexts, i.e., the number of queries to* OChallEnc. *Let* ThFHE *be an $(\ell+q_c, \delta)$-*OW-CPA *secure scheme with $\mathcal{M} = \{0,1\}^\gamma$, such that any adversary $\mathcal{B}$ of circuit size s has advantage $\mathsf{Adv}^{(\ell+q_c,\delta)\text{-OW-CPA}}_{\mathsf{ThFHE}}(\mathcal{B}) \le 2^{-\lambda}$, where $\lambda \ge 3\log_2(1/\varepsilon) + \log_2(2\gamma) + \delta$. Further, we assume that* ThFHE *fulfills (s', ε')-circuit privacy, where $s' = O(s\gamma^3\varepsilon^{-4})$*[6] *and $\varepsilon' = 5q_c\delta\varepsilon/\ell$ and $\varepsilon' = 5q_c\delta\varepsilon/\ell$. Then,* ThFHE$'$ *is ℓ-*IND-CPA *secure with $\mathcal{M}' = \{0,1\}^\delta$; concretely, for any adversary $\mathcal{A}$ of circuit size s' it yields*

$$\mathsf{Adv}^{\ell\text{-IND-CPA}}_{\mathsf{ThFHE}'}(\mathcal{A}) \le \frac{25q_c\delta\varepsilon + 1}{2}.$$

[6] The hidden constant in the $O(\cdot)$ notation is the same as that in the proof of Lemma 1, which can be derived from the Goldreich-Levin theorem.

Choosing the Parameters. To ensure a small enough advantage, since q_c, δ are relatively small, it suffices to choose a small enough ε, which we denote $\varepsilon = 2^{-\lambda'}$. We then require $\lambda = \delta + 3\lambda' + \log_2(2\gamma)$, which determines the required security level of the original OW-CPA scheme. There's therefore a tradeoff between the increased security requirement and the value δ, which improves ciphertext expansion. For instance, if $\lambda' = 128$ then by choosing $\delta = 118, \gamma = 512$, we can pack 118 message bits into each FHE ciphertext, which must encrypt 512 actual bits using the OW-CPA scheme. In this case, to achieve security according to the reduction, the parameters of the OW-CPA scheme would need to be chosen for $\lambda = 512$-bit security. We note that this way of setting parameters may be overly conservative, since our reduction is not tight—unlike with the number of queries ℓ and the matching attack, we are not aware of any weaknesses from choosing smaller values of λ.

Proof. Recall that we are given an OW-CPA secure threshold decryption scheme $\mathsf{ThFHE} = (\mathsf{Setup}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{PartDec}, \mathsf{Combine})$ with message space $\mathcal{M} = \{0,1\}^\gamma$ and we want to construct a new threshold scheme $\mathsf{ThFHE}' = (\mathsf{Setup}', \mathsf{Enc}', \mathsf{Eval}', \mathsf{PartDec}', \mathsf{Combine}')$ with message space $\mathcal{M}' = \{0,1\}^\delta$, which fulfills IND-CPA security. In the IND-CPA security game (Definition 14), the adversary has access to three different oracles, $\mathsf{OEnc}', \mathsf{OChallEnc}'$ and $\mathsf{OPartDec}'$. In the following, we define a sequence of games which modify how the different oracles are implemented. The first game consists of the IND-CPA security game, where $b = 1$. The last game consists of the IND-CPA security game, where $b = 0$.

Game_0:

Queries to OEnc': On input the message $m = (m_i)_{i\in[\delta]} \in \{0,1\}^\delta = \mathcal{M}'$, sample $x, s_1, \ldots, s_\delta \leftarrow U(\mathcal{M})$, compute $c_0 \leftarrow \mathsf{Enc}(\mathsf{pk}, x)$, and set $c_i = \langle x, s_i\rangle + m_i \bmod 2$ for all $i \in [\delta]$. Set $m^{(0)} = m^{(1)} = m$. Output the ciphertext $\mathsf{ct} = (c_0, s_1, \ldots, s_\delta, c_1, \ldots, c_\delta)$ and store $(m^{(0)}, m^{(1)}, \mathsf{ct})$ in the list L.

Queries to $\mathsf{OChallEnc}'$: On input messages $m^{(0)}, m^{(1)} \in \{0,1\}^\delta = \mathcal{M}'$, sample $x, s_1, \ldots, s_\delta \leftarrow U(\mathcal{M})$, compute $c_0 \leftarrow \mathsf{Enc}(\mathsf{pk}, x)$, and set $c_i = \langle x, s_i\rangle + m_i^{(1)}$ for all $i \in [\delta]$. Output $\mathsf{ct} = (c_0, s_1, \ldots, s_\delta, c_1, \ldots, c_\delta)$ and store $(m^{(0)}, m^{(1)}, \mathsf{ct})$ in the list L.

Queries to $\mathsf{OPartDec}'$: On input a circuit C and indices $\iota_1, \ldots, \iota_k$, find the corresponding ciphertexts $\mathsf{ct}_1, \ldots, \mathsf{ct}_k$ in the list L. First compute $\mathsf{ct} \leftarrow \mathsf{Eval}'(\mathsf{pk}, C, \mathsf{ct}_1, \ldots, \mathsf{ct}_k)$ (by internally calling Eval on associated circuit $\widetilde{C}$) and then $d_i \leftarrow \mathsf{PartDec}'(\mathsf{sk}_i, \mathsf{ct})$ (by internally calling PartDec) for all $i \in [n]$. Output $\mathbf{d} = (d_i)_{i\in[n]}$.

Game_1:

Queries to OEnc' and to $\mathsf{OChallEnc}'$: as in Game_0

Queries to $\mathsf{OPartDec}'$: On input a circuit C and indices $\iota_1, \ldots, \iota_k$, find the corresponding messages $m_1^{(1)}, \ldots, m_k^{(1)}$ in the list L. Define the constant circuit $\widetilde{C}$ which, on any input simply outputs $C(m_1^{(1)}, \ldots, m_k^{(1)})$. First compute $\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, \widetilde{C}(x_1^{(1)}, \ldots, x_k^{(1)}))$ (on arbitrary input $x_i^{(1)}$) and then $d_i \leftarrow \mathsf{PartDec}(\mathsf{sk}_i, \mathsf{ct})$ for all $i \in [n]$. Output $\mathbf{d} = (d_i)_{i\in[n]}$.

Game_2:

Queries to OEnc' and to $\mathsf{OPartDec}'$: as in Game_1
Queries to $\mathsf{OChallEnc}'$: On input messages $m^{(0)}, m^{(1)} \in \{0,1\}^\delta = \mathcal{M}'$, sample $x, s_1, \ldots, s_\delta \leftarrow U(\mathcal{M})$ and compute $c_0 \leftarrow \mathsf{Enc}(\mathsf{pk}, x)$. Further, sample $r_1, \ldots, r_\delta \leftarrow U(\{0,1\})$ and set $c_i = r_i + m_i^{(1)} \bmod 2$ for all $i \in [\delta]$. Output $\mathsf{ct} = (c_0, s_1, \ldots, s_\delta, c_1, \ldots, c_\delta)$ and store $(m^{(0)}, m^{(1)}, \mathsf{ct})$ in $\mathtt{L}$.

Game_3:

Queries to OEnc' and to $\mathsf{OPartDec}'$: as in Game_2
Queries to $\mathsf{OChallEnc}'$: On input messages $m^{(0)}, m^{(1)} \in \{0,1\}^\delta = \mathcal{M}'$, sample $x, s_1, \ldots, s_\delta \leftarrow U(\mathcal{M})$ and compute $c_0 \leftarrow \mathsf{Enc}(\mathsf{pk}, x)$. Further, sample $r_1, \ldots, r_\delta \leftarrow U(\{0,1\})$ and set $c_i = r_i + m_i^{(0)} \bmod 2$ for all $i \in [\delta]$. Output $\mathsf{ct} = (c_0, s_1, \ldots, s_\delta, c_1, \ldots, c_\delta)$ and store $(m^{(0)}, m^{(1)}, \mathsf{ct})$ in $\mathtt{L}$.

Game_4:

Queries to OEnc' and to $\mathsf{OPartDec}'$: as in Game_3
Queries to $\mathsf{OChallEnc}'$: On input messages $m^{(0)}, m^{(1)} \in \{0,1\}^\delta = \mathcal{M}'$, sample $x, s_1, \ldots, s_\delta \leftarrow U(\mathcal{M})$, compute $c_0 \leftarrow \mathsf{Enc}(\mathsf{pk}, x)$ and set $c_i = \langle x, s_i \rangle + m_i^{(0)} \bmod 2$ for all $i \in [\delta]$. Output $\mathsf{ct} = (c_0, s_1, \ldots, s_\delta, c_1, \ldots, c_\delta)$ and store $(m^{(0)}, m^{(1)}, \mathsf{ct})$ in $\mathtt{L}$.

Game_5:

Queries to OEnc' and to $\mathsf{OChallEnc}'$: as in Game_4
Queries to $\mathsf{OPartDec}'$: On input a circuit C and indices $\iota_1, \ldots, \iota_k$, find the corresponding ciphertexts $\mathsf{ct}_1, \ldots, \mathsf{ct}_k$ in the list $\mathtt{L}$. First compute $\mathsf{ct} \leftarrow \mathsf{Eval}'(\mathsf{pk}, C, \mathsf{ct}_1, \ldots, \mathsf{ct}_k)$ and then $d_i \leftarrow \mathsf{PartDec}'(\mathsf{sk}_i, \mathsf{ct})$ for all $i \in [n]$. Output $\mathbf{d} = (d_i)_{i \in [n]}$.

Claim. Assume there is an adversary $\mathcal{A}$ of circuit size s' who wins the ℓ-IND-CPA game against ThFHE' with probability at least p. Then, there exists an $i \in \{0, \ldots, 4\}$ such that $\mathsf{cdist}_{s'}(\mathsf{Game}_i, \mathsf{Game}_{i+1}) > (2p-1)/5 := \widetilde{\varepsilon}$.

Next, we argue for all $i \in \{0, \ldots, 4\}$, if $\mathsf{cdist}_{s'}(\mathsf{Game}_i, \mathsf{Game}_{i+1}) > \widetilde{\varepsilon}$, it either breaks one-way security of ThFHE or circuit privacy of ThFHE'. Note that the modifications from Game_0 to Game_1 are the same (in reverse order) as from Game_4 to Game_5. Similarly, the modifications from Game_1 to Game_2 are the same as from Game_3 to Game_4. Moreover, Game_2 and Game_3 are information-theoretically close to each other, because the challenge messages $m^{(0)}$ or $m^{(1)}$ are hidden by truly random bits. We thus focus on the step from Game_0 to Game_1 and the step from Game_1 to Game_2 in the following. The step from Game_0 to Game_1 is necessary to correctly apply the Goldreich-Levin extractor argument in the next step. By replacing the evaluation algorithm with the direct encryption of the evaluated circuit, we make sure that partial decryptions do not leak any information on the challenge bit $b = 1$.[7]

[7] We have overseen this subtlety in an earlier version of this paper and thank the Asiacrypt reviewers for pointing it out to us.

Claim (Game_0 *to* Game_1*).* Assuming that $\mathsf{cdist}_{s'}(\mathsf{Game}_0, \mathsf{Game}_1) > \widetilde{\varepsilon}$ contradicts the $(s', \widetilde{\varepsilon}/\ell)$-circuit privacy of ThFHE'.

Proof. Recall that ℓ denotes the maximal number of allowed queries to $\mathsf{OPartDec}'$. As Game_0 and Game_1 only differ on how queries to $\mathsf{OPartDec}'$ are answered, it yields

$$\widetilde{\varepsilon} < \mathsf{cdist}_{s'}(\mathsf{Game}_0, \mathsf{Game}_1) \leq \ell \cdot \mathsf{cdist}_{s'}(\mathbf{d}, \widetilde{\mathbf{d}}),$$

where $\mathbf{d}$ is a vector of partial decryptions output by the oracle in Game_0 and $\widetilde{\mathbf{d}}$ a vector of partial decryptions output by the oracle in Game_1. Using that applying the randomized function $\mathsf{PartDec}'(\mathsf{sk}_i, \cdot)$ does not increase the computational distance and using the definitions of $\mathsf{PartDec}'$ and Eval' (through $\mathsf{PartDec}$ and Eval, respectively), we observe that $\widetilde{\varepsilon}/\ell$ is bounded above by

$$\mathsf{cdist}_{s'}\left(((\mathsf{sk}_i)_i, \mathsf{Eval}(\mathsf{pk}, \widetilde{C}, \mathsf{ct}_1, \ldots, \mathsf{ct}_k)), ((\mathsf{sk}_i)_i, \mathsf{Enc}(\mathsf{pk}, \widetilde{C}(x_1^{(1)}, \ldots, x_k^{(1)})))\right),$$

contradicting the $(s', \widetilde{\varepsilon}/\ell)$-circuit privacy of ThFHE (cf. Definition 12). We later link $\widetilde{\varepsilon}$ to ε' as in the theorem statement. ■

Claim (Game_1 *to* Game_2*).* Assuming that $\mathsf{cdist}_{s'}(\mathsf{Game}_1, \mathsf{Game}_2) > \widetilde{\varepsilon}$ contradicts the $(\ell + q_c, \delta)$-$\mathsf{OW\text{-}CPA}$ security assumption of ThFHE.

Proof. Let q_c denote the number of allowed queries to $\mathsf{OChallEnc}'$. As Game_1 and Game_2 only differ on how queries to $\mathsf{OChallEnc}'$ are answered, it yields

$$\widetilde{\varepsilon} < \mathsf{cdist}_{s'}(\mathsf{Game}_1, \mathsf{Game}_2) \leq q_c \cdot \mathsf{cdist}_{s'}(\mathsf{ct}, \widetilde{\mathsf{ct}}),$$

where ct is an encryption output by the oracle in Game_1 and $\widetilde{\mathsf{ct}}$ is an encryption output by the oracle in Game_2. We can rewrite ct using the concatenated Goldreich-Levin extractor $\mathcal{E}$ from Definition 2. We define $m := (0, \ldots, 0, m^{(1)}) \in \mathcal{M}^\delta \times \mathcal{M}'$, $X := U(\mathcal{M})$ and $Z := (\mathsf{Enc}(\mathsf{pk}, X), (\mathsf{sk}_i)_{i \in S}, \mathtt{E}, \mathtt{PartD})$, the latter being the random variable defined by the randomized encryption algorithm for uniform random messages, the corrupted secret key shares, the circuit evaluations and the partial decryptions given by the partial decryption queries in the security game. Furthermore, we set $Y = X$, such that $\mathsf{cdist}_{s'}((X, Z), (Y, Z)) \leq \varepsilon$ for all $s', \varepsilon > 0$. We observe that $\mathsf{ct} = (Z, \mathcal{E}(X, U(M^\delta) + m)$ and $\widetilde{\mathsf{ct}} = (Z, U(\mathcal{M}^\delta \times \mathcal{M}') + m)$. It holds that

$$\widetilde{\varepsilon}/q_c < \mathsf{cdist}_{s'}(\mathsf{ct}, \widetilde{\mathsf{ct}}) \leq \mathsf{cdist}_{s'}\left((Z, \mathcal{E}(X, U(M^\delta)), (Z, U(\mathcal{M}^\delta \times \mathcal{M}'))\right).$$

Applying Lemma 1 implies an upper bound on the unpredictability entropy, i.e., $H^{\mathsf{unp}}_{\varepsilon,s}(X|Z) < \lambda$, where $\varepsilon = \frac{\widetilde{\varepsilon}}{5q_c\delta}$, $s = O(s'\gamma^{-3}\varepsilon^4)$ and $\lambda = \delta + \log_2 2\gamma + 3\log_2 1/\varepsilon$. To conclude the proof of the claim, we link the unpredictability entropy of X given Z to the $\mathsf{OW\text{-}CPA}$ security of ThFHE via a reduction. In the following, we explain how the corresponding oracle queries for ThFHE' (which define X and Z) can be answered by having access to the three analogue oracles (denoted $\mathsf{OEnc}, \mathsf{OChallEnc}$ and $\mathsf{OPartDec}$) from the $\mathsf{OW\text{-}CPA}$ security game, cf. Definition 13.

Reduction to $\mathsf{OW\text{-}CPA}$ *Game.*

Queries to OEnc': On input the message $m = (m_i)_{i\in[\delta]} \in \{0,1\}^\delta = \mathcal{M}'$, sample $x, s_1, \ldots, s_\delta \leftarrow U(\mathcal{M})$ and query OEnc on input x. Take the received $c_0 = \mathsf{Enc}(\mathsf{pk}, x)$ and compute $c_i = \langle x, s_i\rangle + m_i \bmod 2$ for all $i \in [\delta]$. Set $m^{(0)} = m^{(1)} = m$. Output the ciphertext $\mathsf{ct} = (c_0, s_1, \ldots, s_\delta, c_1, \ldots, c_\delta)$ and store $(m^{(0)}, m^{(1)}, \mathsf{ct})$ in the list $\mathtt{L}$.

Queries to $\mathsf{OChallEnc}'$: On input $m^{(0)}, m^{(1)} \in \{0,1\}^\delta = \mathcal{M}'$, query $\mathsf{OChallEnc}$ (on no input) and get back an encryption $c_0 = \mathsf{Enc}(\mathsf{pk}, x)$ for an unknown x. For $i \in [\delta]$, sample $s_i \leftarrow U(\mathcal{M})$. Define the circuit $\tilde{C}$ which takes as input x and computes $\langle x, s_i\rangle + m_i^{(1)} \bmod 2$ for every $i \in [\delta]$. Then query $\mathsf{OPartDec}$ on c_0 and the circuit $\tilde{C}$. For every $i \in [\delta]$, the partial decryption oracle outputs all partial decryption shares that can be combined to $c_i = \langle x, s_i\rangle + m_i^{(1)}$. Output the ciphertext $\mathsf{ct} = (c_0, s_1, \ldots, s_\delta, c_1, \ldots, c_\delta)$ and store $(m^{(0)}, m^{(1)}, \mathsf{ct})$ in $\mathtt{L}$.

Queries to $\mathsf{OPartDec}'$: On input a circuit C and indices $\iota_1, \ldots, \iota_k$, find the corresponding messages $m_1^{(1)}, \ldots, m_k^{(1)}$ in the list $\mathtt{L}$. Define the constant circuit $\widetilde{C}$ which, on any input, simply outputs $C(m_1^{(1)}, \ldots, m_k^{(1)})$. Query $\mathsf{OPartDec}$ on the circuit $\widetilde{C}$ and the indices $\iota_1, \ldots, \iota_k$ as input. On output $\mathbf{d} = (d_i)_{i\in[n]}$ of the oracle $\mathsf{OPartDec}$, output $\mathbf{d}$.

Queries to $\mathsf{OPartDec}$ done within $\mathsf{OChallEnc}'$ and within $\mathsf{OPartDec}'$ do pass the entropy-check with the entropy bound δ (cf. line 10 of Fig. 2). Regarding the first case, by Lemma 2, every inner product $\langle x, s_i\rangle \bmod 2$ leaks at most one bit of x. Hence, at most δ bits are leaked in total when querying $\mathsf{OPartDec}$ on circuit $\tilde{C}$. A similar argument holds for $\mathsf{OPartDec}'$: The circuit C' leaks at most δ bits for every x_i with $i \in [k]$. Note that every query to $\mathsf{OChallEnc}'$ leads to one query to $\mathsf{OPartDec}$. Similarly, every query to $\mathsf{OPartDec}'$ leads to one query to $\mathsf{OPartDec}$. Thus, the $\mathsf{OW\text{-}CPA}$ scheme has to allow for $\ell + q_c$ partial decryption queries in total. To conclude the proof, we observe that $H^{\mathsf{unp}}_{\varepsilon,s}(X|Z) < \lambda$ implies that for any adversary $\mathcal{B}$ of circuit size s

$$2^{-\lambda} < \Pr[\mathcal{B}(Z) = X] \leq \mathsf{Adv}^{(\ell+q_c,\delta)\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{ThFHE}}(\mathcal{B}).$$

■

Regarding the parameters from the theorem statement, we observe from the above two sub proofs that $\varepsilon' = \widetilde{\varepsilon}/\ell = \varepsilon 5q_c\delta/\ell$ and $s' = O(s\gamma^3\varepsilon^{-4})$ as stated. □

5 Threshold Fully Homomorphic Encryption from LWE with Polynomial Modulus

We now present our construction of a t-out-of-n ThFHE scheme with $\mathsf{OW\text{-}CPA}$ security. First, we describe and analyze our main construction based on any LSSS with strong $\{0,1\}$-reconstruction. Then, in Sect. 5.5, we give an alternative construction that combines pseudorandom secret sharing with Shamir sharing to improve efficiency when $\binom{n}{t}$ is small.

By applying the $\mathsf{OW\text{-}CPA}$ to $\mathsf{IND\text{-}CPA}$ transformation for ThFHE from Sect. 4, we hence obtain an $\mathsf{IND\text{-}CPA}$ secure scheme. When we restrict ourselves to

standard PKE, our construction gives us a standard ThPKE scheme (cf. Definition 10). We can then also apply the alternative transformation which additionally achieves some form of robustness.

5.1 Nearly Linear Decryption of FHE

We use the following abstraction of LWE-based encryption schemes, where decryption is viewed as a linear function of the secret key that outputs a "noisy" version of the correct message. Similar notions were used in [BKS19, Bra+19].

Definition 15 (FHE with (β, ε)-linear decryption). *Let* $\mathsf{FHE} := (\mathsf{Setup}, \mathsf{Enc}, \mathsf{Dec}, \mathsf{Eval})$ *be a fully homomorphic encryption scheme (as in Definition 11) with message space* $\mathcal{M} \subseteq R_p$ *and ciphertext space* R_q^r*. Suppose that* Setup *outputs a secret key* $\mathsf{sk} \in R_q^r$ *which has the form* $(1, \mathbf{s})$ *for some* $\mathbf{s} \in R_q^{r-1}$*.*

Let $\beta = \beta(\lambda) \in \mathbb{N}, \varepsilon = \varepsilon(\lambda) \in [0, 1]$*. We say that* FHE *has* (β, ε)-linear decryption *if for any* $\lambda, \kappa \in \mathbb{N}$*,* $(\mathsf{pp}, \mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Setup}(1^\lambda, 1^\kappa)$*, depth-*$\kappa$ *circuit* $C \colon \mathcal{M}^k \to \mathcal{M}$*, messages* $m_1, \ldots, m_k \in R_p$*, ciphertexts* $\mathbf{c}_i \leftarrow \mathsf{Enc}(\mathsf{pk}, m_i) \in R_q^r$ *and* $\mathsf{ct} \leftarrow \mathsf{Eval}(\mathsf{pk}, \mathbf{c}_1, \ldots, \mathbf{c}_k)$*, it holds that*

$$\langle \mathsf{sk}, \mathsf{ct} \rangle = \lfloor q/p \cdot C(m_1, \ldots, m_k) \rceil + e \mod q,$$

for some $e \in R_q$ *such that* $\Pr[\|e\|_\infty \leq \beta] \geq 1 - \varepsilon$ *(where the probability is taken over the randomness of* Setup, Enc *and* Eval*).*

In standard (Module)-LWE based constructions, it's possible to securely set the parameters such that the ratio β/q can be made arbitrarily small, and as long as we have $\beta/q = 1/\mathsf{poly}(\lambda)$, then q is $\mathsf{poly}(\lambda)$.

For security, we require that FHE is IND-CPA secure.[8] This can be instantiated under the Module-LWE assumption to obtain (leveled) FHE using, for instance, the BGV scheme [BGV12] (with superpolynomial q). For $p = 2$, $d = 1$ and $R = \mathbb{Z}$, we also get (leveled) FHE under the standard LWE assumption with a polynomial modulus q [BV14].

5.2 Construction from LSSS with Strong $\{0, 1\}$-Reconstruction

Our construction works over the ring $R = \mathbb{Z}[X]/f(X)$ for some degree-d irreducible polynomial f, and uses the following main ingredients:

- $\mathcal{D}_{\mathsf{flood}}$: a noise distribution over $\mathbb{Z}_q$ with magnitude bounded by β_{flood},
- $\mathcal{D}_{\mathsf{sim}}$: a noise distribution over $\mathbb{Z}_q$, where $\mathrm{RD}_a(\mathcal{D}_{\mathsf{sim}} \| \mathcal{D}_{\mathsf{flood}} + B) \leq \varepsilon_{\mathrm{RD}_a}$, for some $a \in (1, \infty), \varepsilon_{\mathrm{RD}_a} > 1$ and for all B with $|B| \leq \beta_{\mathsf{fhe}}$,
- LSS: a t-out-of-n linear secret sharing scheme $\mathsf{LSS} = (\mathsf{Share}, (\mathsf{Rec}_S)_{S \subset [n]})$ with strong $\{0, 1\}$-reconstruction, associated parameters $L, \tau_{\mathsf{max}}, \tau_{\mathsf{min}}$ and shares in $\mathbb{Z}_q^L$ (cf. Definition 6),

[8] In our main construction, we assume $\mathcal{M}$ is large and only rely on OW-CPA security of FHE. When extending to smaller $\mathcal{M}$ in Sect. 5.3, we instead need IND-CPA security.

– FHE: a OW-CPA secure FHE $= (\mathsf{Setup}', \mathsf{Enc}, \mathsf{Eval}, \mathsf{Dec})$ scheme with message space $\mathcal{M} \subseteq R_p$, ciphertext space R_q^r, and $(\beta_{\mathsf{fhe}}, \varepsilon)$-linear decryption for some $\beta_{\mathsf{fhe}} < q/(2p) - \tau_{\mathsf{min}}\beta_{\mathsf{flood}}$ and some negligible ε.

We now define the scheme ThFHE $:= (\mathsf{Setup}, \mathsf{Enc}, \mathsf{Eval}, \mathsf{PartDec}, \mathsf{Combine})$ by using Enc and Eval from the underlying FHE scheme and setting Setup, PartDec and Combine as specified in Fig. 3. We prove its correctness in the full version [BS23].

For now, we assume the plaintext space $\mathcal{M} \subseteq R_p$ is superpolynomial in the security parameter, so that FHE is OW-CPA secure. In Sect. 5.3, we show how to extend this to use FHE with any plaintext space, which allows instantiating from LWE with polynomial modulus.

We write $\mathcal{D}_{\mathsf{flood},R_q^r}$ (resp. $\mathcal{D}_{\mathsf{sim},R_q^r}$) to refer to the distribution consisting of rd independent $\mathcal{D}_{\mathsf{flood}}$ (resp. $\mathcal{D}_{\mathsf{sim}}$) random variables, used to sample the coefficients of r elements of R_q.

We show security in the following.

Theorem 2 (Security). *For any adversary $\mathcal{A}$ against the (ℓ, ν)-OW-CPA property of the ThFHE scheme in Fig. 3 with message space $\mathcal{M}$, there exists an adversary $\mathcal{B}$ against the IND-CPA property of FHE, such that*

$$\mathsf{Adv}^{(\ell,\nu)\text{-OW-CPA}}_{\mathsf{ThFHE}}(\mathcal{A}) \leq \left[|\mathtt{ChallM}| \left(\mathsf{Adv}^{\mathsf{IND\text{-}CPA}}_{\mathsf{FHE}}(\mathcal{B}) + 2^{-\log_2(|\mathcal{M}|)+\nu}\right) \cdot \varepsilon_{\mathrm{RD}_a}^{\ell d(nL-\tau_{\mathsf{max}})}\right]^{(a-1)/a} + \ell\varepsilon,$$

where L and τ_{max} are parameters from the LSS and $|\mathtt{ChallM}|$ is the number of challenge ciphertexts the adversary queried.

Proof. The high-level idea is to modify the (ℓ, ν)-OW-CPA game (Fig. 2) such that the t secret shares and the answers to the ℓ partial decryption queries provided to the adversary no longer depend on the underlying secret key sk. This is reflected by the sequence of games from G_0 to G_4. In the new game G_4,

$\mathsf{Setup}(1^\lambda, 1^\kappa, n, t)$

1 : $(\mathsf{pp}, \mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{Setup}'(1^\lambda, 1^\kappa)$
2 : // $\mathsf{sk} \in R_q^r, \mathsf{sk}_i \in (R_q^r)^L$
3 : $(\mathsf{sk}_1, \ldots, \mathsf{sk}_n) \leftarrow \mathsf{LSS.Share}(\mathsf{sk})$
4 : **return** $(\mathsf{pp}, \mathsf{pk}, \mathsf{sk}_1, \ldots, \mathsf{sk}_n)$

$\mathsf{PartDec}(\mathsf{sk}_i, \mathsf{ct})$

1 : $\mathbf{e}_{i,j} \leftarrow \mathcal{D}_{\mathsf{flood},R_q}$ for $j \in [L]$
2 : // $\mathsf{sk}_i = (\mathsf{sk}_{i,1}, \ldots, \mathsf{sk}_{i,L}) \in (R_q^r)^L$
3 : $\mathbf{d}_{i,j} \leftarrow \langle \mathsf{ct}, \mathsf{sk}_{i,j} \rangle + \mathbf{e}_{i,j}$
4 : **return** $\mathbf{d}_i \leftarrow (\mathbf{d}_{i,1}, \ldots, \mathbf{d}_{i,L})$

$\mathsf{Combine}(\{\mathbf{d}_i\}_{i \in S}, \mathsf{ct})$

1 : $y \leftarrow \mathsf{Rec}_S((\mathbf{d}_i)_{i \in S})$
2 : **return** $\lfloor (p/q) \cdot y \rceil$

Fig. 3. Setup, partial decrypt and combine algorithms for OW-CPA secure ThFHE. The Enc and Eval algorithms are the same as for FHE.

the adversary still learns the circuit evaluations, stored in the set $\mathtt{E}$, which might leak some information on the challenge messages, stored in $\mathtt{ChallM}$. In a final step, when going to G_5, we make those circuit evaluations independent of the challenge ciphertexts, by tweaking the oracle $\mathsf{OChallEnc}$ to output random ciphertexts (independent of the challenge messages). Here we need to assume the $\mathsf{IND\text{-}CPA}$ security of the underlying non-threshold FHE scheme. By arguing that the circuit evaluations coming from the partial decryption queries do not leak too much information on the challenge messages, we can bound the advantage of the resulting adversary in the last game G_5 to be negligible.

Game G_0**:** This is the real threshold (ℓ, ν)-$\mathsf{OW\text{-}CPA}$ experiment as in Fig. 2. The view of $\mathcal{A}$ is given by

$$\mathcal{V} = (\mathsf{pp}, \mathsf{pk}, \{\mathsf{sk}_i\}_{i\in S}, \mathtt{CT}, \mathtt{ChallCT}, \mathtt{E}, \mathtt{PartD}),$$

where pp are the public parameters, pk is the public key, $\{\mathsf{sk}_i\}_{i\in S}$ are the secret key shares given to the adversary, $\mathtt{CT}$ and $\mathtt{ChallCT}$ contain the (challenge) ciphertexts the adversary has queried, $\mathtt{E}$ and $\mathtt{PartD}$ store the results of up to ℓ adaptive circuit evaluations and partial decryption queries. In each partial decryption query, $\mathcal{A}$ inputs a circuit C and list of indices $(i_1, \ldots, i_k)$, and receives $(\mathbf{d}_i)_{i\in[n]}$, where $\mathbf{d}_i$ is the partial decryption of $\mathsf{ct} \leftarrow \mathsf{Eval}(\mathsf{pk}, C, \mathsf{ct}_{i_1}, \ldots, \mathsf{ct}_{i_k})$ under sk_i. Once the adversary knows all the partial decryption shares, they can reconstruct the circuit evaluation $C(m_{i_1}, \ldots, m_{i_k})$. It yields, $\mathsf{Adv}^{(\ell,\nu)\text{-}\mathsf{OW\text{-}CPA}}_{\mathsf{ThFHE}}(\mathcal{A}) = \mathsf{Adv}^{G_0}_{\mathsf{ThFHE}}(\mathcal{A})$.

Game G_1**:** In this game, we redefine how the partial decryptions are computed. After the adversary chooses the set $S \subset [n]$ of corrupt parties, let $S_L = \{(i,j)\}_{i\in S, j\in[L]}$ be the corresponding set of share elements. Fix $T \supseteq S_L$ to be a maximal invalid set of share elements. Then, compute the partial decryptions $\mathbf{d}_i$ for a ciphertext ct as follows:

1. For $(i,j) \in T$, let $\tilde{\mathbf{d}}_{i,j} = \langle \mathsf{ct}, \mathsf{sk}_{i,j}\rangle$;
2. For $(i,j) \in ([n] \times [L]) \setminus T$, let $T_{i,j} \subseteq T \cup \{(i,j)\}$ be a minimal valid set of share elements, and compute $\tilde{\mathbf{d}}_{i,j} = \langle \mathsf{ct}, \mathsf{sk}\rangle - \sum_{(k,l)\in T_{i,j}\setminus\{(i,j)\}} \tilde{\mathbf{d}}_{k,l}$;
3. Sample $\mathbf{e}_i \leftarrow \mathcal{D}_{\mathsf{flood}, R_q^L}$ and compute $\mathbf{d}_i = \tilde{\mathbf{d}}_i + \mathbf{e}_i$, for $i \in [n]$.

Game G_2**:** In this game, before outputting the partial decryptions for a ciphertext ct, we first check that $\langle \mathsf{ct}, \mathsf{sk}\rangle = \lfloor q/p \rceil \cdot C(m_1, \ldots, m_k) + e$ for some e with $\|e\|_\infty \leq \beta_{\mathsf{fhe}}$. If not, the game aborts.

Game G_3**:** We replace the partial decryptions corresponding to shares outside of T with simulated ones. Firstly, in step (2) above, for $(i,j) \in ([n] \times [L]) \setminus T$, we now compute $\tilde{\mathbf{d}}_{i,j}$ as $\tilde{\mathbf{d}}_{i,j} = \lfloor q/p \cdot C(m_1, \ldots, m_k) \rceil - \sum_{(k,l)\in T_{i,j}\setminus\{(i,j)\}} \tilde{\mathbf{d}}_{k,l}$. Secondly, in step (3), instead of always sampling $\mathbf{e}_{i,j} \leftarrow \mathcal{D}_{\mathsf{flood}, R_q}$, we only sample $\mathbf{e}_{i,j} \leftarrow \mathcal{D}_{\mathsf{flood}, R_q}$ if $(i,j) \in T$, and $\mathbf{e}_{i,j} \leftarrow \mathcal{D}_{\mathsf{sim}, R_q}$ otherwise.

Game G_4**.** In the next game, we change how the secret key shares are sampled: pick $(\mathsf{sk}'_1, \ldots, \mathsf{sk}'_n) \leftarrow \mathsf{LSS.Share}(0)$ and give to $\mathcal{A}$ the shares $\{\mathsf{sk}'_i\}_{i\in S}$.

Game G_5. In the last game, we replace the oracle $\mathsf{OChallEnc}$ by $\mathsf{OChallEnc}'$, as defined in Fig. 4. In the new oracle, two independent m and m' are sampled. Whereas m is added to the challenge message list $\mathtt{ChallM}$, the encryption of m' is added to the challenge ciphertext list $\mathtt{ChallCT}$.

$\mathsf{OChallEnc}'()$

1 : $\mathsf{idx} = \mathsf{idx} + 1$
2 : $m, m' \leftarrow \mathcal{M}$
3 : $\mathtt{ChallM} = \mathtt{ChallM} \cup \{m\}$
4 : $\mathsf{ct} \leftarrow \mathsf{Enc}(\mathsf{pk}, m')$
5 : $\mathtt{ChallCT} = \mathtt{ChallCT} \cup \{\mathsf{ct}\}$
6 : $\mathtt{L}[\mathsf{idx}] := \{(1, m, \mathsf{ct})\}$
7 : **return** ct

Fig. 4. Modified $\mathsf{OChallEnc}'$ oracle.

The theorem then follows from the following lemmata relating the advantages between the different games and showing that the final advantage in the last game is negligibly small.

Lemma 6. *For any* PPT *adversary* $\mathcal{A}$ *in Games* G_0 *and* G_1*, it holds that*

$$\mathsf{Adv}^{G_0}_{\mathsf{ThFHE}}(\mathcal{A}) = \mathsf{Adv}^{G_1}_{\mathsf{ThFHE}}(\mathcal{A}).$$

Proof. Note that the view of $\mathcal{A}$ in G_1 is identical to that in G_0, due to the strong $\{0,1\}$-reconstruction property of LSS. This is because every share belonging to the maximally invalid set T is computed the same way as in G_0, using the shares sk_i, while each share outside this set is deterministically fixed to be a sharing of the correct secret $\langle \mathsf{ct}, \mathsf{sk} \rangle$, plus noise sampled from $\mathcal{D}_{\mathsf{flood}}$, as in G_0. Hence, $\mathsf{Adv}^{G_0}_{\mathsf{ThFHE}}(\mathcal{A}) = \mathsf{Adv}^{G_1}_{\mathsf{ThFHE}}(\mathcal{A})$. ■

Lemma 7. *For any* PPT *adversary* $\mathcal{A}$ *in Games* G_1 *and* G_2*, it holds that*

$$\mathsf{Adv}^{G_1}_{\mathsf{ThFHE}}(\mathcal{A}) \leq \mathsf{Adv}^{G_2}_{\mathsf{ThFHE}}(\mathcal{A}) + \ell\varepsilon.$$

Proof. Due to the $(\beta_{\mathsf{fhe}}, \varepsilon)$-linear decryption property of FHE, and applying a union bound over the ℓ queries, we have that $\mathsf{Adv}^{G_1}_{\mathsf{ThFHE}}(\mathcal{A}) \leq \mathsf{Adv}^{G_2}_{\mathsf{ThFHE}}(\mathcal{A}) + \ell\varepsilon$. ■

Lemma 8. *For any* PPT *adversary* $\mathcal{A}$ *in Games* G_2 *and* G_3*, it holds that*

$$\mathsf{Adv}^{G_2}_{\mathsf{ThFHE}}(\mathcal{A}) \leq (\mathsf{Adv}^{G_3}_{\mathsf{ThFHE}}(\mathcal{A}) \cdot \varepsilon_{\mathrm{RD}_a}^{\ell d(nL-\tau_{\mathsf{max}})})^{(a-1)/a},$$

where τ_{max} *is the size of the smallest maximal invalid share set in* LSS*.*

Proof. We compute the Rényi divergence between the views of the adversary in each game. Each view consists of the adversary's random tape and

$$\mathcal{V} = (\mathsf{pp}, \mathsf{pk}, \{\mathsf{sk}_i\}_{i \in S}, \mathtt{CT}, \mathtt{ChallCT}, \mathtt{E}, \mathtt{PartD}),$$

where $\mathtt{CT}$ and $\mathtt{ChallCT}$ store the (challenge) ciphertexts and $\mathtt{E}$ and $\mathtt{PartD}$ the circuit evaluations and partial decryption shares after the ℓ partial decryption queries. For simpler notation, we set $\mathbf{i}^\eta := (i_1^\eta, \ldots, i_k^\eta)$ and $\mathbf{m}^\eta := (m_{i_1}^\eta, \ldots, m_{i_k}^\eta)$ for the index list and corresponding message vector of the η-th query. Let D_2 and D_3 denote the distributions of $\mathcal{V}$ in games G_2 and G_3, respectively. Since the partial decryption queries are adaptive, note that the circuit C^η and the index list $\mathbf{i}^\eta$ input during the η-th query depend on the previous queries to $\mathsf{OEnc}, \mathsf{OChallEnc}$ and $\mathsf{OPartDec}$ and the corresponding responses. However, since each $(C^\eta, \mathbf{i}^\eta)$ is a deterministic function of the other values in the view (including the random tape), by the data processing inequality (Lemma 3), $\mathrm{RD}_a(D_2 \| D_3) \leq \mathrm{RD}_a(D_2' \| D_3')$, where D_2', D_3' are the distributions with the $C^\eta, \mathbf{i}^\eta$ values removed. D_2' are D_3' are now defined identically, except in the way the partial decryption components $\mathbf{d}_{i,j}^\eta$ are computed for indices $(i,j) \notin T$. In G_2, $\mathbf{d}_{i,j}^\eta$ is computed using (amongst other values) $\langle \mathsf{ct}^\eta, \mathsf{sk} \rangle + \mathcal{D}_{\mathsf{flood}, R_q}$, whereas G_3 instead uses $\lfloor q/p \cdot C(\mathbf{m}^\eta) \rceil + \mathcal{D}_{\mathsf{sim}, R_q}$. Since $\langle \mathsf{ct}^\eta, \mathsf{sk} \rangle = \lfloor q/p \cdot C(\mathbf{m}^\eta) \rceil + e_\eta$ for some e_η with $\|e_\eta\|_\infty \leq \beta_{\mathsf{fhe}}$, and the view contains $nL - |T|$ pairs $(i,j) \notin T$ where the sampling of $\mathbf{d}_{i,j}^\eta$ changes from G_2 to G_3, to compute $\mathrm{RD}_a(D_2' \| D_3')$, it suffices to compute

$$\mathrm{RD}_a\left(((e_1 + \mathcal{D}_{\mathsf{flood}, R_q})^{nL-|T|}, \ldots, (e_\ell + \mathcal{D}_{\mathsf{flood}, R_q})^{nL-|T|}) \| \mathcal{D}_{\mathsf{sim}, R_q}^{\ell(nL-|T|)}\right).$$

Applying Lemma 5 with $N = d\ell(nL - |T|)$, $D_1 = \mathcal{D}_{\mathsf{flood}}, D_2 = \mathcal{D}_{\mathsf{sim}}$, we get

$$\mathrm{RD}_a(D_2' \| D_3') \leq \varepsilon_{\mathrm{RD}_a}^{d\ell(nL-|T|)}.$$

Applying the probability preservation property of Rényi divergence, we bound the success probability of the adversary as required. ∎

Lemma 9. *For any* PPT *adversary* $\mathcal{A}$ *in Games* G_3 *and* G_4*, it holds that*

$$\mathsf{Adv}_{\mathsf{ThFHE}}^{G_3}(\mathcal{A}) = \mathsf{Adv}_{\mathsf{ThFHE}}^{G_4}(\mathcal{A}).$$

Proof. Note that the view of $\mathcal{A}$ in G_4 is perfectly indistinguishable from the one in G_3 by the perfect privacy property of LSS. Hence, $\mathsf{Adv}_{\mathsf{ThFHE}}^{G_3}(\mathcal{A}) = \mathsf{Adv}_{\mathsf{ThFHE}}^{G_4}(\mathcal{A})$. ∎

Lemma 10. *For any* PPT *adversary* $\mathcal{A}$ *in Games* G_4 *and* G_5*, it holds that*

$$\mathsf{Adv}_{\mathsf{ThFHE}}^{G_4}(\mathcal{A}) \leq \mathsf{Adv}_{\mathsf{ThFHE}}^{G_5}(\mathcal{A}) + |\mathtt{ChallM}| \cdot \mathsf{Adv}_{\mathsf{FHE}}^{\mathsf{IND\text{-}CPA}}(\mathcal{A}).$$

Proof. The view of $\mathcal{A}$ in G_5 and G_4 are computationally indistinguishable assuming the IND-CPA security of the non-threshold FHE scheme for every query to $\mathsf{OChallEnc}'$. In total, there are $|\mathtt{ChallM}|$ many such queries. Hence, we obtain $\mathsf{Adv}_{\mathsf{ThFHE}}^{G_4}(\mathcal{A}) \leq \mathsf{Adv}_{\mathsf{ThFHE}}^{G_5}(\mathcal{A}) + |\mathtt{ChallM}| \cdot \mathsf{Adv}_{\mathsf{FHE}}^{\mathsf{IND\text{-}CPA}}(\mathcal{A})$. ∎

Lemma 11. *For any* PPT *adversary* $\mathcal{A}$ *in Game* G_5*, it yields that*

$$\mathsf{Adv}^{G_5}_{\mathsf{ThFHE}}(\mathcal{A}) \leq q_c \cdot 2^{-\log_2(|\mathcal{M}|)} \cdot 2^{\nu},$$

where $\mathcal{M}$ *is the message space,* ν *the bound on the entropy leakage guaranteed in the* (ℓ, ν)-OW-CPA *game and* $q_c := |\mathtt{ChallM}|$ *the number of queried ciphertext challenges. If* ν *is logarithmic,* $|\mathcal{M}|$ *exponential and* q_c *polynomial in* λ*, the advantage is negligible in* λ*.*

Proof. Let $\mathcal{V}$ denote the views of $\mathcal{A}$ in Game G_5. It is given by

$$\mathcal{V} = (\mathsf{pp}, \mathsf{pk}, \{\mathsf{sk}_i\}_{i \in S}, \mathtt{CT}, \mathtt{ChallCT}, \mathtt{E}, \mathtt{PartD}) .$$

Note that in Game G_5, all challenge messages in `ChallM` are independent of the challenge ciphertexts in `ChallCT`. Furthermore, the secret key shares $\{\mathsf{sk}_i\}_{i \in S}$ are independent of the secret key sk and hence also independent of the challenge messages in `ChallM`. The same is true for the simulated partial decryption shares stored in `PartD`. The public parameters pp, public key pk and normal ciphertexts stored in `CT`, are trivially independent of `ChallM`. Thus, $\widetilde{H}_\infty(m|\mathcal{V}) = \widetilde{H}_\infty(m|\mathtt{E})$. Overall, it yields

$$\begin{aligned}\mathsf{Adv}^{G_5}_{\mathsf{ThFHE}}(\mathcal{A}) &\leq \sum_{m \in \mathtt{ChallM}} 2^{-\widetilde{H}_\infty(m|\mathcal{V})} = \sum_{m \in \mathtt{ChallM}} 2^{-\widetilde{H}_\infty(m|\mathtt{E})} \\ &\leq \sum_{m \in \mathtt{ChallM}} 2^{-\widetilde{H}_\infty(m)+\nu},\end{aligned}$$

where we used that the leakage is guaranteed to be bounded above by ν. Finally, we use that every $m \in \mathtt{ChallM}$ is sampled uniformly at random over $\mathcal{M}$, thus $\widetilde{H}_\infty(m) = \log_2(|\mathcal{M}|)$, leading to $\mathsf{Adv}^{G_5}_{\mathsf{ThFHE}}(\mathcal{A}) \leq |\mathtt{ChallM}| \cdot 2^{-\log_2(|\mathcal{M}|)} \cdot 2^{\nu}$. ■

□

5.3 Supporting a Larger Plaintext Space

The above construction works for a plaintext space $\mathcal{M} \subseteq R_p$. Since we only obtain one-way security, this requires $|R_p|$ to be superpolynomial in λ to give a meaningful security guarantee. If R_p is small, we can easily modify our threshold scheme to still be secure by using several ciphertexts to encrypt larger messages with the underlying FHE scheme. Note that this change is necessary to obtain an instantiation from LWE with polynomial modulus, since there $\mathcal{M} = R_p = \mathbb{Z}_2$.

Concretely, suppose that FHE is IND-CPA secure and has small message space $\mathcal{M}$. Define FHE' with message space $\mathcal{M}^k$, such that $\left|\mathcal{M}^{-k}\right|$ is negligible, by encrypting each of the k message components separately under FHE. We then instantiate our threshold scheme using FHE' instead of FHE, where during the partial decrypt and combine steps, we run the algorithms for the previous construction on each component separately. If FHE is IND-CPA secure, then so is FHE', and the proof carries over in the same way, except that the ℓ values in the statement of Theorem 2 will be replaced with $k\ell$, to account for the fact that each of the ℓ decryption queries involves k decryptions of ciphertexts from FHE.

5.4 Bounding the Rényi Divergence

We now analyze parameters and instantiate the distributions $\mathcal{D}_{\mathsf{flood}}$ and $\mathcal{D}_{\mathsf{sim}}$. For now, we simply choose them both to be rounded Gaussian distributions $\lfloor D_\sigma \rceil$ with the same standard deviation σ. In Sect. 6.1, we obtain tighter parameters by carefully optimizing the choice of distributions. If FHE has a maximum ciphertext noise bound of β_{fhe}, then using Lemma 4 with our choice of distributions, we get $\varepsilon_{\mathrm{RD}_a} = \mathrm{RD}_a(\mathcal{D}_{\mathsf{flood}} + \beta_{\mathsf{fhe}} \| \mathcal{D}_{\mathsf{sim}}) \leq \exp\left(\frac{a\beta_{\mathsf{fhe}}^2}{2\sigma^2}\right)$. If FHE has λ_{FHE} bits of security, then from Theorem 2, the resulting ThFHE scheme is λ_{ThFHE}-bit secure, such that

$$\lambda_{\mathsf{ThFHE}} \geq (\lambda_{\mathsf{FHE}} - \ell d(nL - \tau_{\mathsf{max}}) \log_2 \varepsilon_{\mathrm{RD}_a}) \frac{a-1}{a} \tag{1}$$

Combining the above two equations, we obtain $\lambda_{\mathsf{ThFHE}} \geq \frac{a-1}{a}\lambda_{\mathsf{FHE}} - \ell d(nL - \tau_{\mathsf{max}})(a-1)\frac{\beta_{\mathsf{fhe}}^2}{2\sigma^2}\log_2 e$. Setting for instance $a = \lambda_{\mathsf{ThFHE}}$, and choosing $\sigma, q, \beta_{\mathsf{fhe}}$ such that $\sigma = O(\beta_{\mathsf{fhe}}\sqrt{\ell d(nL - \tau_{\mathsf{max}})(a-1)})$ while decryption is still correct, the loss in security is only a constant factor. Smaller values of a give different tradeoffs between the size of σ and the security loss. Note that in any case, if ℓ and nL are polynomially bounded then both σ and the modulus q can be also.

5.5 Alternative Construction Using Pseudorandom Secret Sharing

In the full version of this paper [BS23], we also give a different construction based on pseudorandom secret sharing (PRSS), which improves upon the previous one in some aspects. Instead of having each party perturb their share by an independent, random noise term, we will use PRSS [GI99, CDI05]. This allows them to jointly sample replicated secret sharings of small noise terms, without interaction, after a one-time setup that distributes PRF keys. We also exploit the fact that replicated secret shares can be locally converted to any other LSS, and convert the secret shared noise terms into Shamir sharings before using them for partial decryption. This means that the partial decryptions are Shamir shares, which are much smaller, consisting of only 1 element over R_q each. Furthermore, this leads to improved parameters in the security reduction (by avoiding the $nL - \tau_{\mathsf{max}}$ term in Eq. 1), and we can additionally take advantage of the error-correction capability of Shamir to achieve strong robustness when $t < n/3$. This offers a way of getting robustness for ThFHE instead of only ThPKE with our previous transformations, with the drawback that we require $\binom{n}{t}$ to be not too large, due to using replicated secret sharing.

To sum up, PRSS is a lightweight tool for achieving robustness with a small number of parties.

6 Sample Parameters and Security Estimates

In this section, we discuss how to choose concrete parameters for our OW-CPA secure threshold construction, where we take as a starting point the lattice-based

scheme Kyber [Sch+20]. Hence, we are not in the fully homomorphic case, but in the standard PKE case and thus obtain a standard ThPKE scheme. We denote the thresholdized version of Kyber by TKyber.

The relevant parameters for Kyber are the ring degree d, the rank r, the modulus q and the width η of the secret key and encryption randomness distributions. Whereas the specifications of Kyber only consider three parameter sets, called Kyber512, Kyber768 and Kyber1024, we additionally consider three more parameter sets, that we subsequently call Kyber1280, Kyber1536 and Kyber1792. As the name suggest, they are obtained in a similar manner as the previous parameter sets, simply by increasing the rank by +1.

6.1 Security from the Reduction

Let λ_{PKE} (resp. λ_{ThPKE}) denote the security level of the starting PKE (resp. the resulting ThPKE) from Theorem 2. Further, we set $\Delta_\lambda := \lambda_{\mathsf{PKE}} - \lambda_{\mathsf{ThPKE}}$, which describe the security loss in our reduction. Instantiating Eq. 1 in the standard PKE setting yields

$$\lambda_{\mathsf{ThPKE}} \geq \frac{a-1}{a} \cdot \left(\lambda_{\mathsf{PKE}} - \ell d(nL - \tau_{\mathsf{max}}) \log_2 \varepsilon_{\mathrm{RD}_a}\right), \tag{2}$$

where ℓ is the number of partial decryption queries, d the degree of the ring R, L and τ_{max} parameters of the underlying LSSS and $\varepsilon_{\mathrm{RD}_a}$ an upper bound on the Rényi divergence $\mathrm{RD}_a(\mathcal{D}_{\mathsf{sim}} \| \mathcal{D}_{\mathsf{flood}} + \beta_{\mathsf{pke}})$ of order a. Here, $\mathcal{D}_{\mathsf{sim}}$ (resp. $\mathcal{D}_{\mathsf{flood}}$) denotes the simulating (resp. flooding) noise distribution and β_{pke} is a bound on the decryption noise that depends on the concrete parameters of Kyber, in particular on the ring degree d, the module rank r and the parameter η, as well as the maximal failure probability ε we want to achieve. For concreteness we set λ_{PKE} as the core-SVP classical hardness, i.e., the resulting BKZ block estimated from the Lattice Estimator [APS15] size multiplied by 0.292.

Table 2 and Table 3 present some sample parameters. The relevant difference between the two is that in the first table, we focus on larger numbers of parties n and samples ℓ while accepting a modulus of up to 39 bits. For simplicity, we assume that both $\mathcal{D}_{\mathsf{flood}}$ and $\mathcal{D}_{\mathsf{sim}}$ follow a Gaussian distribution of width σ.

Table 2. Sample parameters and security estimates following the reduction from Theorem 2 using a generic approach.

Set	$(\beta_{\mathsf{pke}}, \varepsilon)$	n	t	ℓ	$\lceil \log_2 \sigma \rceil$	$\lceil \log_2 q \rceil$	λ_{PKE}	λ_{ThPKE}	Δ_λ
TKyber1024	$(390, 2^{-60})$	2	1	1	17	23	120	117	3
TKyber1024	$(934, 2^{-300})$	2	1	1	18	24	111	108	3
TKyber1024	$(390, 2^{-60})$	10	9	1	17	25	105	102	3
TKyber1280	$(435, 2^{-60})$	10	5	1	21	29	120	117	3
TKyber1536	$(476, 2^{-60})$	20	10	10	27	36	112	109	3
TKyber1792	$(513, 2^{-60})$	2	1	2^{32}	33	39	123	120	3

Table 3. Sample parameters and security estimates following the reduction from Theorem 2 obtained from a hand-tuned Python program.

Set	q	n	t	ℓ	$\mathcal{D}_{\mathsf{flood}}$	$\mathcal{D}_{\mathsf{sim}}$	λ_{ThPKE}	Δ_λ
TKyber1024	$5 \cdot 3329$	2	1	1	947	1087	100	111
TKyber1024	$10 \cdot 3329$	2	1	2	1994	2034	104	91
TKyber1024	$9 \cdot 3329$	3	2	1	1197	1297	106	92

In contrast, in the second table we fine-tuned the flooding and simulation distributions so that we can allow for very small q (only multiplying the original Kyber modulus by small constants up to 10).

Acknowledgments. We would like to thank Ivan Damgård and Aayush Jain for helpful discussions on secret sharing, and the anonymous reviewers for pointing out a flaw in an earlier version of this work, and other valuable feedback.

This work has been supported by the Independent Research Fund Denmark (DFF) under project number 0165-00107B (C3PO), the Protocol Labs Research Grant Program RFP-013 and the Aarhus University Research Foundation.

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